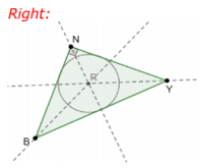


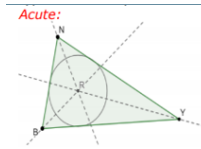
Unit 11 Overview: Justifying Relationships in Polygons With Circles

Unit Narrative



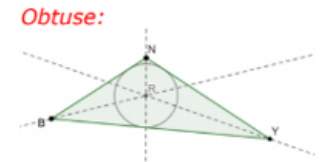
In this unit, students initially explore how to circumscribe and inscribe triangles in a circle, and then discover relationships, postulates, and theorems associated with triangles. Students are aware of some of the relationships, postulates, and theorems from their work with triangles during middle school, and will extend this surface-level knowledge as they perform constructions and write proofs to develop a deeper understanding of these relationships, postulates, and theorems.

In Section 1, students focus on circumscribing and inscribing triangles and quadrilaterals in circles. In Lesson 1, students explore how to construct the circumscribed circle of a triangle and locate the circumcenter of a triangle. Then, in Lesson 2, they discover how to construct an inscribed circle in a triangle and locate the incenter of a triangle. Students solve mathematical problems involving triangles and quadrilaterals inscribed in a circle in Lesson 3. This section concludes with Lesson 4, where students solve mathematical problems involving quadrilaterals inscribed in a circle. Lesson H1, where students construct an equilateral triangle, square, and regular hexagon inscribed in a circle, is designed to be taught after Lesson 2 in Honors Geometry courses.



In Section 2, students explore a variety of relationships, postulates, and theorems, and apply their knowledge and skills to solve mathematical and real-world problems. In Lesson 5, students show that the sum of the interior angles of a triangle is 180° , and the sum of the exterior angles of a triangle is 360° . Then, in Lesson 6, students determine if a set of triangle side lengths can be used to construct a triangle and use the triangle inequality theorem to prove relationships in triangles. Next, in

Lesson 7, students find the centroid of a triangle and use the relationship of the distance a centroid is from a vertex to the midpoint of the opposite side to solve problems. Subsequently, in Lesson 8, students discover the triangle midsegment theorem using AA similarity, and use the theorem to prove relationships and to solve problems. Finally, in Lesson 9, students apply what they've learned in lessons 5-8 to solve mathematical and real-world problems.



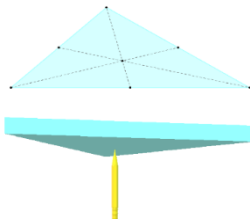
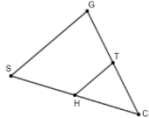
Unit At-A-Glance

Section 1: Circumscribing and Inscribing	Benchmarks Addressed	Lesson 1	Constructing a Circumscribed Circle of a Triangle
	MA.912.GR.5.3 MA.912.GR.6.3	Lesson 2	Constructing the Incenter of a Triangle
		Lesson H1	Inscribing Polygons in Circles
		Lesson 3	Solving Problems Involving Triangles Inscribed in Circles
		Lesson 4	Solving Problems Involving Quadrilaterals Inscribed in Circles
Section 2: Triangle Relationships, Postulates, and Theorems	Benchmarks Addressed	Lesson 5	Interior and Exterior Angles of Triangles
	MA.912.GR.1.3 MA.912.GR.5.2 MA.912.LT.4.10 MA.912.GR.5.4 (Honors only)	Lesson 6	Triangle Inequality Theorem
		Lesson 7	Centroid of a Triangle
		Lesson 8	Triangle Midsegment Theorem
		Lesson 9	Solving Problems Using Triangle Relationships



Differentiation

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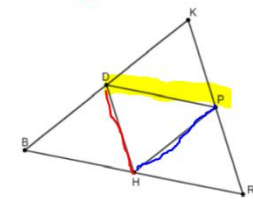
Engagement The “WHY” of learning	Consider opportunities to recruit interest, sustain effort, and self-regulate for increased motivation. - In 11.3.1 Warm-Up, students investigate a number pattern displayed on rocks arranged in a triangle. Students first pose questions about the pattern and then describe the pattern. Solving non-routine problems, such as this puzzle, can help students build strategic competence. - In 11.5.1 Warm-Up, students share and justify the superpowers of a geometry superhero and villain. These prompts are intended to inspire student curiosity and interest. This component can help build students’ agency and mathematical identities as each and every student can participate. 
Representation The “WHAT” of learning	Offer differentiated entry points for engaging with content in a variety of ways.  - Provide kinesthetic and visual entry points by giving students triangular manipulatives (made from cardstock/cardboard or triangular snack chips, such as Doritos) in 11.7.2. Exploration. Prompt students to find the balancing point with their pencils. Connect to students’ prior experiences balancing on a see saw. - Provide a kinesthetic entry point by structuring 11.9.2 Activity as a circuit. Students’ discussions during the activity also provide an auditory entry point. Finally, provide a visual entry point by displaying the relationships, postulates, and theorems from the unit on classroom Word Walls and/or anchor charts for students to reference during the activity.
Action & Expression The “HOW” of learning	Consider providing different methods for students to demonstrate and express their learning. - Consider allowing students to use GeoGebra to explore the relationships, postulates, and theorems during 11.9.2 Activity. - In 11.8.3 Guided Practice, consider providing students with a bank of words to choose from to complete the proof in question 3. Explain that words in the bank can be used more than once. Also, include words that are not used in the word bank. Prompt students to read and rewrite the proof several times to help them internalize the concepts.  3. Triangles SGC and HTC are shown. Given: $SH = HC$ and $GT = TC$ Prove: $SG \parallel HT$ Since $SH = HC$ and $GT = TC$, point H is the <u>midpoint</u> of line segment SC and point T is the <u>midpoint</u> of line segment GC by the <u>definition of midpoint</u>. Therefore, HT is the <u>midsegment</u> of $\triangle SGC$ by definition and $SG \parallel HT$ by the <u>triangle midsegment theorem</u>.

ELL Information

Help students develop an understanding of the key terms and additional glossary in the unit by connecting the new words to words they are familiar with, and associating the words with visual representations. For example:

- When introducing the word “midsegment” in 11.8.2 Exploration, underline the prefix, “mid-,” and prompt students to make connections to prior vocabulary, such as “midpoint” and “middle.”
- Be judicious with the use of different colored pencils or markers. For example, prompt students to color code the opposite angles in the diagrams in 11.4.3 Guided Practice.
- In 11.8.3 Guided Practice, use corresponding colors to label variable expressions and the side lengths or angles that they represent.

Triangle BKR is shown, where D , P , and H are the midpoints and the perimeter of $\triangle DPH$ is 23.5. $DP = 17x$, $PH = (6x + 5)$, and $HD = 20x - 3$.



All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support” below:

- Circumscribed circle
- Inscribed circle
- Inscribed polygon in a circle

Additional Differentiated Support

Scaffolding Resources	On-Ramp
In 11.1.2 Exploration, students are asked to write conjectures about the locations of right, acute, and obtuse triangles inscribed in circles. Then, students discuss similarities and differences among their conjectures. To support students in this process, it may be helpful to provide them with sentence starters that they can build around until they get comfortable with creating sentences on their own. For example: “All of the triangles_____.” “Only the right triangle_____.” “Only the obtuse triangle_____.” “Only the acute triangle_____.” For students who need more assistance with creating sentences, you can provide them with sentence choices to choose from. They can verbally explain what they mean by the option that they choose, so that you can confirm if they are correct. For example, in 11.4.5 Wrap-Up, where students are asked to identify and correct an error in a sample of student work: A) Anastasia made the angles congruent. They should be supplementary. B) Anastasia made the angles congruent. They should be complementary.	Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish. • Area and Circumference of a Circle • Similar Triangles

Anastasia's Work
$3I + 23 = 6I + 13$ $3I = 10$ $I = \frac{10}{3}$



Section Overview

Section 1: Circumscribing and Inscribing (Lessons 1-4)

Benchmarks Addressed	Learning Targets	
MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle. <i>Clarification 1: Instruction includes using a compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.</i> MA.912.GR.5.4 (Honors only) Construct a regular polygon inscribed in a circle. Regular polygons are limited to triangles, quadrilaterals, and hexagons. <i>Clarification 1: When given a circle, the center must be provided.</i> <i>Clarification 2: Instruction includes using a compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.</i> MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle. <i>Clarification 1: Instruction includes triangles in a circle and semicircle.</i>	Lesson 1	- I can construct the circumscribed circle of a triangle. - I can locate the circumcenter of a triangle.
	Lesson 2	- I can construct an inscribed circle in a triangle. - I can locate the incenter of a triangle.
	Lesson H1	- I can construct an equilateral triangle inscribed in a circle. - I can construct a square inscribed in a circle. - I can construct a regular hexagon in a circle.
	Lesson 3	I can solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.
	Lesson 4	I can solve mathematical problems involving quadrilaterals inscribed in a circle.

Section 2: Triangle Relationships, Postulates, and Theorems (Lessons 5-9 & H1)

Benchmarks Addressed	Learning Targets	
MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships and theorems of triangles. <i>Clarification 1: Postulates, relationships and theorems include measures of interior angles of a triangle sum to 180°; measures of a set of exterior angles of a triangle sum to 360°; triangle inequality theorem; base angles of isosceles triangles are</i>	Lesson 5	- I can show that the sum of the interior angles of a triangle is 180 degrees. - I can show that the sum of the exterior angles of a triangle is 360 degrees.

<i>congruent; the segment joining midpoints of two sides of a triangle is parallel to the third side and half the length; the medians of a triangle meet at a point.</i> <i>Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.</i> <i>Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.</i> MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment. <i>Clarification 1: Instruction includes using compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.</i> MA.912.GR.5.4 (Honors only) Construct a regular polygon inscribed in a circle. Regular polygons are limited to triangles, quadrilaterals, and hexagons. <i>Clarification 1: When given a circle, the center must be provided.</i> <i>Clarification 2: Instruction includes using a compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.</i> MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements. <i>Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.</i>	Lesson 6	- I can determine if a set of triangle side lengths can be used to construct a triangle. - I can use the triangle inequality theorem to prove relationships in triangles.
	Lesson 7	- I can find the centroid of a triangle. - I can use the relationship of the distance a centroid is from a vertex to the midpoint of the opposite side to solve problems.
	Lesson 8	- I can discover the triangle midsegment theorem using AA similarity. - I can use the triangle midsegment theorem to prove relationships and to solve problems.
	Lesson 9	I can solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.



Vertical Alignment

8th Grade Unit 13
Properties and Theorems of
Triangles



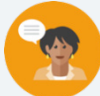
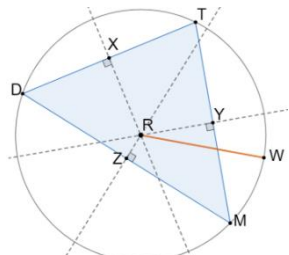
Geometry Unit 11
Triangles

Prior Course(s)	Course Level
MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be found from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be found from a given set of sides. MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles. MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.	MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment. MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle. MA.912.GR.6.3 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle. MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles. MA.912.GR.5.4 (<i>Honors only</i>) Construct a regular polygon inscribed in a circle. Regular polygons are limited to triangles, quadrilaterals, and hexagons. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

11.1 Constructing a Circumscribed Circle of a Triangle

Lesson Introduction

 Benchmark Focus	MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.		 Learning Targets	- I can construct the circumscribed circle of a triangle. - I can locate the circumcenter of a triangle.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How do you construct perpendicular bisectors of each side of a triangle? - How do you construct the circumscribed circle of a triangle? - How do you locate the circumcenter of a triangle?			
 Lesson Narrative	In this lesson, students discover that they can construct the circumscribed circle of a triangle by first constructing the perpendicular bisectors of the triangle and identifying the circumcenter. They discover that the circumcenter is equidistant from each of the vertices of the triangle. Then, they explore how they can place a compass on the circumcenter, adjust the length to reach each vertex of the triangle, and complete the circle.			
 Component Summary	11.1.1 Warm-Up (~5 min.)	Compare surface area and volume of lakes (<i>SE p. 177</i>)	11.1.2 Exploration (35-40 min.)	Discover how to construct the circumscribed circle of a triangle (<i>SE p. 178</i>)
	11.1.3 Wrap-Up (~5 min.)	Demonstrate mastery of lesson learning targets (<i>SE p. 181</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Inscribed Polygon in a Circle - Circumscribed Circle - Point of Concurrency - Circumcenter <u>Additional Glossary:</u> - Perpendicular Bisector		- Patty Paper - Compass - Straight Edge	
	<u>Additional Preparation</u> Prior to the lesson, work through 11.1.2 Exploration to ensure that you are familiar with the steps students will take and are prepared to support them as they work.			

 Teacher Tips	Common Misconceptions - Students may not understand that if a point is on the perpendicular bisector of a segment, then it is equidistant from the endpoints of the segment. - Students may not understand why the intersection of the perpendicular bisectors, the circumcenter, is equidistant from each vertex of the triangle. - Students may not understand how to use a compass and straightedge to construct perpendicular bisectors.	Things to Consider - In 11.1.2 Exploration, consider modeling the constructions and then having students practice independently a few times as they develop fluency with skills. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate when the circumcenter is inside/outside the triangle and on a side of the triangle.
	Literacy and Language Routines Compare and Connect In 11.1.2 Exploration, foster students’ meta-awareness by prompting them to identify, compare, and contrast the locations of circumcenters in acute, obtuse, and right triangles. Support students’ meta-cognitive and meta-linguistic awareness as they share what the similarities and differences that they notice in the diagrams.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.1.1: Participate in Effortful Learning 11.1.2 Exploration is a long and challenging activity. Foster perseverance in students by modeling the steps needed to perform the constructions, and recognizing their efforts during the activity. Develop students’ abilities to analyze and solve the problems by using the think aloud strategy to demonstrate expectations for student responses to the problems.
	 Differentiate Instruction	In 11.1.2 Exploration, students are provided with a kinesthetic entry point as they use patty paper, compasses, and straightedges to construct perpendicular bisectors, a circumcenter, and a circle. Students are also provided with an auditory entry point as they discuss similarities and differences among the right, obtuse, and acute inscribed triangles. Provide a visual entry point by showing students images and definitions of obtuse, acute, and right triangles. 
Lesson Notes:		

11.1.1 Warm-Up: Lake Okeechobee and Deep Lake

Component Overview: Students compare the surface area and volume of Lake Okeechobee and Deep Lake to determine why it takes more Deep Lakes to cover the surface of Lake Okeechobee than to fill up Lake Okeechobee.



11.1.1 Warm-Up: Lake Okeechobee and Deep Lake

Lake Okeechobee means "big water" in the language of the Seminole tribe. Lake Okeechobee has a surface area of 730 square miles. Deep Lake in Big Cypress National Preserve is a naturally occurring sinkhole that was formed roughly 6,000 years ago. Deep Lake has a surface area of 0.002 square miles.

1. How many Deep Lakes are needed to cover the same surface area as Lake Okeechobee?

365,000 lakes

Lake Okeechobee averages 9 feet in depth, whereas Deep Lake averages 95 feet in depth. Based on the volume of the two lakes it would take approximately 30,850 Deep Lakes to fill up Lake Okeechobee.

2. Explain why it takes more Deep Lakes to cover the surface area of Lake Okeechobee than to fill up Lake Okeechobee.

Answers will vary. Surface area refers to the area of the water at the surface of the area (two-dimensional), while the volume includes the third dimension of height (in this case, depth). It takes more Deep Lakes to cover the surface of Lake Okeechobee than to fill it because Deep Lake is narrower but much deeper than Lake Okeechobee.



Consider sharing images, videos, or three-dimensional models to help students conceptualize the difference between surface area and volume. For example, you could show students a long thin graduated cylinder and a petri dish.

11.1.2 Exploration: Circumscribing A Triangle With A Circle

Component Overview: Students explore how to circumscribe a triangle with a circle through a series of scaffolded problems they complete with a partner. Students will need a compass, straightedge, and patty paper. Circulate while students work, to support struggling students and determine which students you will call on during the debrief discussion.



11.1.2 Exploration: Circumscribing A Triangle with A Circle



An **inscribed polygon in a circle** is a polygon which has all of its vertices on a circle.

$\triangle TWD$ is inscribed in Circle R , where $\overline{RW}$ is a radius. Lines $\overline{RX}$, $\overline{RY}$, and $\overline{RZ}$ are perpendicular bisectors of $\overline{TD}$, $\overline{TW}$, and $\overline{DW}$, respectively.

Each diagram shows a different type of triangle inscribed in Circle R .



Consider modeling the constructions and then having students practice independently a few times as they develop fluency with skills.

Acute	
Right	
Obtuse	

11.1.2 Exploration: Circumscribing A Triangle With A Circle (continued)

1. Determine similarities and differences among all three inscribed triangles.

Similarities	Differences
<i>Answers will vary. In all triangles, the three perpendicular bisectors intersect at one point.</i>	<i>Answers will vary. The perpendicular bisectors intersect in different places relative to the triangle (inside, on, outside the triangle)</i>

2. Point R is the center of the circle that is circumscribed about $\triangle TWD$. Make a conjecture about point R as it relates to the perpendicular bisectors of the sides of $\triangle TWD$.
Answers will vary. Point R is the intersection of the perpendicular bisectors.
3. Ask two other pairs of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of Their Conjecture	Initials
<i>Answers will vary.</i>	
<i>Answers will vary.</i>	

4. Review your conjecture with your partner and if necessary, make any changes.



A **circumscribed circle** is the smallest circle that includes a plane figure. If the figure is a polygon, then the circle must contain all of the vertices of the polygon. Not every polygon has a circumscribed circle, but all triangles and all regular polygons have circumscribed circles.

The **point of concurrency** is the point where three or more lines intersect.

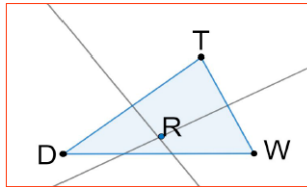


Compare and Connect

Foster students' meta-awareness by prompting them to identify, compare, and contrast the locations of circumcenters in acute, obtuse, and right triangles. Support students' meta-cognitive and meta-linguistic awareness as they share the similarities and differences they have noticed in the diagrams.

11.1.2 Exploration: Circumscribing A Triangle With A Circle (continued)

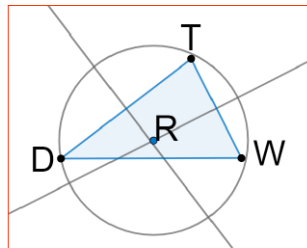
5. Copy $\triangle DTW$ onto patty paper. Construct the perpendicular bisectors of each side. Name the point of concurrency of $\triangle DTW$ point R .



6. Discuss with your partner the relationship point R has with each of the vertices.
Answers will vary. The point seems to be equidistant from each of the vertices.

The point of concurrency of the perpendicular bisectors of a triangle (the center of a triangle's circumscribed circle) is the **circumcenter**.

7. Use the definition of a circumcenter to explain how the relationship point R has with each of the vertices ensures that a circle can be drawn to circumscribe the triangle.
Answers will vary. Because each point is equidistant from the circumcenter, it will ensure that each vertex of the triangle will be on the constructed circle. The distance between the circumcenter and each vertex of the triangle is the radius of the circle.
8. Use a compass centered at point R to draw the circle that will pass through all three vertices.



Students are provided with a kinesthetic entry point as they use patty paper, compasses, and straightedges to construct perpendicular bisectors, a circumcenter, and a circle. Students are also provided with an auditory entry point as they discuss similarities and differences among the right, obtuse, and acute inscribed triangles. Provide a visual entry point by showing students images and definitions of obtuse, acute, and right triangles.



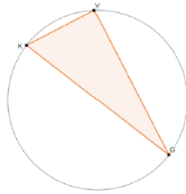
This Exploration is a long and challenging activity. Foster perseverance in students by modeling the steps needed to perform the constructions and recognizing their efforts during the activity. Develop students' abilities to analyze and solve the problems by using the think aloud strategy to demonstrate expectations for student responses to the problems.

11.1.2 Exploration: Circumscribing A Triangle With A Circle (continued)

9. Review your conjecture. Does your conjecture hold true for newly constructed circumscribed Circle R ?

Answers will vary.

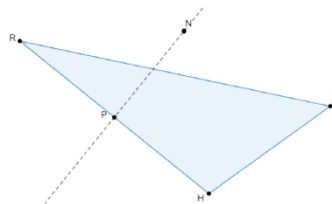
10. Copy $\triangle KYG$ onto another piece of paper. Then, use a compass and a straight edge to construct the circle that circumscribes $\triangle KYG$.



11. Describe how the circumscribed circle can be used to determine the type of triangle that $\triangle KYG$ is. Name a property and theorem from Unit 10 that can be used to verify this.

The circumcenter is the midpoint of the hypotenuse, the inscribed angle that corresponds to the arc intersected by the endpoints would be $\frac{1}{2}$ the arc measure or 90° . Therefore, the triangle is a right triangle by the Inscribed Angle Theorem.

12. The dotted line is the perpendicular bisector of side $\overline{RH}$. The distance between points R and N is 4.35 units. If point N is the circumcenter, what is the distance between points H and N ? Explain or show your reasoning.



If point N is the circumcenter, then $HN = 4.35$ units because the vertices of the triangles will be equidistant from the circumcenter.



Consider using a dynamic tool, such as GeoGebra, to enable students to investigate when the circumcenter is inside/outside the triangle and on a side of the triangle.



While circulating, watch for student responses to question 9. Their answers will provide insight into their understanding of questions 1-8. If necessary, debrief question 9 with a whole-class discussion.



After question 11, students will begin applying what they learned up until this point. Consider having a whole-class discussion after question 11 to ensure students are ready for the subsequent questions.

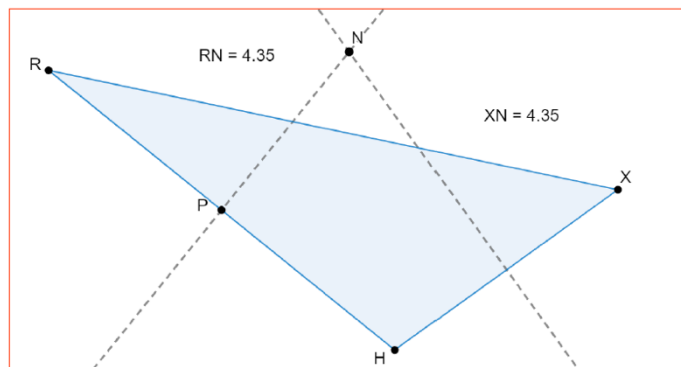


Students may have difficulty with question 12. Students may not label the 90° angles on the perpendicular bisectors. Consider prompting students who are stuck to articulate what information they know about triangle PNH .

11.1.2 Exploration: Circumscribing A Triangle With A Circle (continued)

13. Emilio tells his partner, Talia, that in order to show that point N is the circumcenter, he only needs to construct one more perpendicular bisector. Talia challenged him to prove it. Describe how Emilio could justify that point N is the circumcenter by drawing only one more perpendicular bisector.

If Emilio constructs another perpendicular bisector through $\overline{NX}$ (or $\overline{RX}$), he can determine if N is the circumcenter. If the second perpendicular bisector intersects with point N , then point N is the circumcenter.



11.1.3 Wrap-Up: Ticket Out the Door

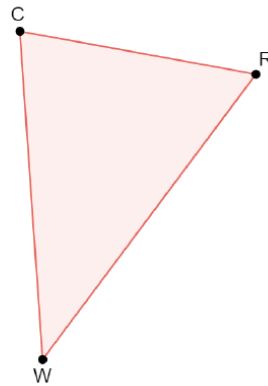
Component Overview: Students describe how to construct a circle that inscribes a triangle.



11.1.3 Wrap-Up: Ticket Out the Door

1. Describe how to construct Circle H , the circumscribed circle of $\triangle CRW$.

Construct perpendicular bisectors of two of the sides of $\triangle CRW$ and label the point of intersection as the circumcenter. Place compass on the circumcenter point and adjust the length to reach any vertex of the triangle. Then, use the compass to complete the circle that passes through all three vertices.

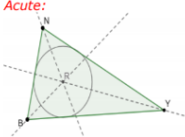
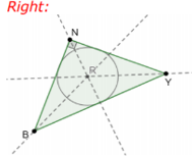
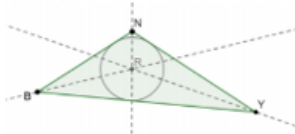


Use data from this formative assessment to determine which students need remediation before the next lesson.

11.2 Constructing the Incenter of a Triangle

Lesson Introduction

 Benchmark Focus	MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.		 Learning Targets	- I can construct an inscribed circle in a triangle. - I can locate the incenter of a triangle.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- How do you construct an inscribed circle in a triangle? - How do you locate the incenter of a triangle?			
 Lesson Narrative	In this lesson, students discover that the point of concurrency of the angle bisectors is the incenter. Then, they construct perpendicular line segments from the incenter to each side of the triangle. Finally, students construct circles inscribed in triangles and compare the locations of the incenter in acute, right, and obtuse triangles.			
 Component Summary	11.2.1 Warm-Up (~5 min.)	Notice and Wonder about triangular pyramid salt piles (<i>SE p. 182</i>)	11.2.2 Exploration (35-40 min.)	Constructing an inscribed circle exploration (<i>SE p. 183</i>)
	11.2.3 Wrap-Up (~5 min.)	Reflect on most difficult or confusing part of the lesson (<i>SE p. 186</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Incenter <u>Additional Glossary:</u> - Angle Bisector - Point of Concurrency		- Compass - Patty Paper - Straightedge	- Prior to the lesson, work through 11.2.2 Exploration to ensure that you are familiar with the steps students will take and are prepared to support them as they work. - Consider creating, discussing, and displaying a Frayer Model to help students distinguish between the terms “incenter” and “circumcenter.”

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not understand how to construct angle bisectors. - Students may not understand how and when to construct an incenter versus a circumcenter. - Students may not recognize that in an equilateral triangle, the incenter and the circumcenter are the same point.	- In 11.2.2 Exploration, consider modeling the constructions and then having students practice independently a few times as they develop fluency with skills. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the location of the incenter in right, acute, and obtuse triangles.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 11.2.1 Warm-Up, students are shown photographs of salt poured onto a triangle until it piles up. Ask students “What do you notice? What do you wonder?” Give students a few minutes to write down things they notice and things they wonder. After students have had a chance to write down their responses, asks several students to share things they noticed and wondered. Record these for all to see. Steer the conversation to wondering about the location of the peak of the salt piles.	MA.K12.MTR.1.1: Participation in Effortful Learning 11.2.2 Exploration is a long and challenging activity. Foster perseverance in students by modeling the steps needed to perform the constructions and recognizing students’ efforts during the activity. Develop students’ abilities to analyze and solve the problems by using the think aloud strategy to demonstrate expectations for student responses to the problems.
 Differentiate Instruction	In 11.2.2 Exploration, students are provided with a kinesthetic entry point as they use patty paper, compasses, and straightedges to construct angle bisectors, an incenter, and a circle. Students are also provided with an auditory entry point as they discuss similarities and differences among the right, obtuse, and acute inscribed triangles. Provide a visual entry point by showing students images and definitions of obtuse, acute, and right triangles.	
	 <i>Acute:</i>  <i>Right:</i>  <i>Obtuse:</i>	
Lesson Notes:		

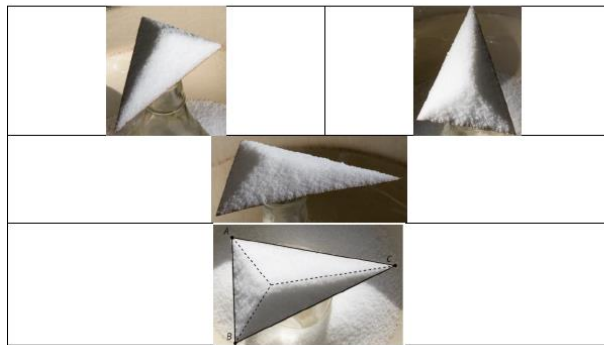
11.2.1 Warm-Up: Salt Pile

Component Overview: Students share what they notice and wonder about photographs of triangular pyramid salt piles.



11.2.1 Warm-Up: Salt Pile

Salt is poured onto a triangle until it piles up. As the salt piles up, the pile reaches a maximum height. As the salt reaches a maximum height, new grains of salt fall off whichever side of the triangle is closest.



Students are shown photographs of salt poured onto a triangle until it piles up. Ask students “What do you notice? What do you wonder?” Give students a few minutes to write down things they notice and things they wonder. After students have had a chance to write down their responses, asks several students to share things they noticed and wondered. Record these for all to see. Steer the conversation to wondering about the location of the peak of the salt piles.

- What do you wonder about the salt pile?
Answers will vary. I wonder how much salt is on the triangle. I wonder how much salt has fallen off the triangle.
- What do you notice about the salt pile?
Answers will vary. I notice that the salt has a point higher than all the others. I notice there is less salt that has fallen off on the back side (AB).
- What do you notice about the point where the salt reaches a maximum?
Answers will vary. I notice that there is more salt at the maximum point than the rest of the triangle. I notice that the maximum point seems to be the same distance from each of the sides.

11.2.2 Exploration: Constructing an Inscribed Circle

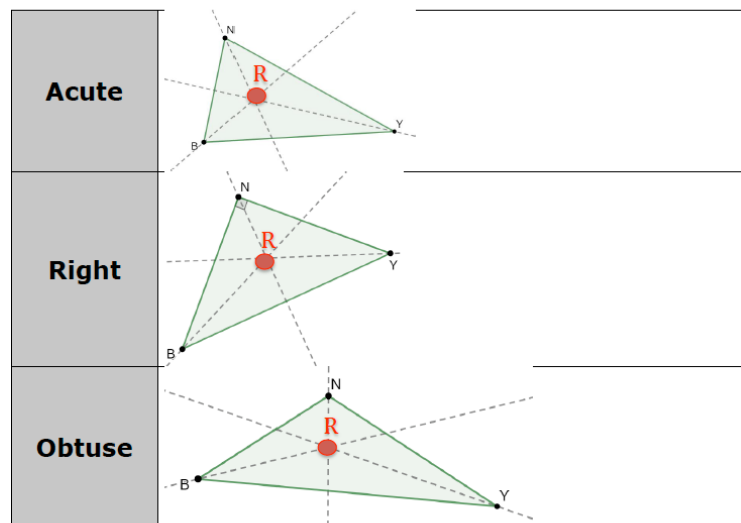
Component Overview: Students discover that the point of concurrency of the angle bisectors is the incenter of the triangle. Then, they construct perpendicular line segments from the incenter to each side of the triangle. Finally, they construct circles inscribed in triangle and compare the locations of the incenters in acute, right, and obtuse triangles.



11.2.2 Exploration: Constructing an Inscribed Circle

1. $\triangle NYB$ is shown as three different types of triangles.

a. Label each point of concurrency as point R .



b. What do you notice about the three different triangles and point R , the point of concurrency?

Answers will vary. I notice that the point R is inside each of the triangles.



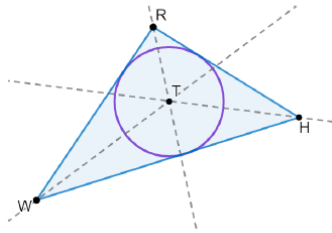
Consider modeling the constructions and then having students practice independently a few times as they develop fluency with skills.



- What do you notice about the point of concurrency of the angle bisectors in right, obtuse, and acute triangles?
- How does this compare to the point of concurrency of the perpendicular bisectors, or circumcenter, in right, obtuse, and acute triangles explored in Lesson 1?

11.2.2 Exploration: Constructing an Inscribed Circle (continued)

2. $\triangle RHW$ is shown where each angle is bisected. The three angle bisectors meet at point T .

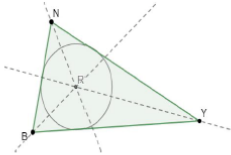


How is circle T related to $\triangle RHW$?

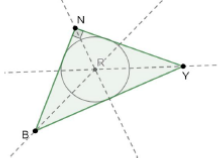
The center of circle T is the point of concurrency of the angle bisectors. The circle is completely inside the triangle.

3. Use a compass to draw the inscribed circles for each type of $\triangle NYB$ in problem 1.

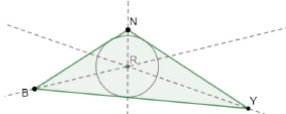
Acute:



Right:



Obtuse:



The point of concurrency of the angle bisectors of a triangle is the **incenter**.



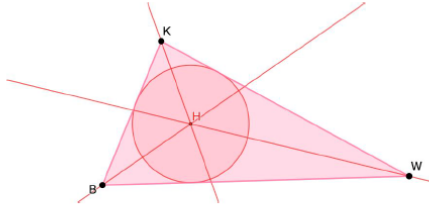
Encourage students to use precise mathematical terms in their written descriptions and verbal discourse. For example, in question 2, highlight students who refer to point T as the “point of concurrency” of the “angle bisectors.” Displaying relevant key terms and an additional glossary on a classroom Word Wall will help students incorporate the new terms into their vocabulary.



Students are provided with a kinesthetic entry point as they use patty paper, compasses, and straightedges to construct angle bisectors, an incenter, and a circle. Students are also provided with an auditory entry point as they discuss similarities and differences among the right, obtuse, and acute inscribed triangles. Provide a visual entry point by showing students images and definitions of obtuse, acute, and right triangles.

11.2.2 Exploration: Constructing an Inscribed Circle (continued)

4. Consider $\triangle KWB$.



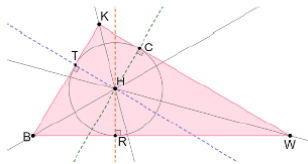
- a. Copy $\triangle KWB$ onto patty paper and find the incenter of the triangle by constructing the angle bisectors. Label the incenter of $\triangle KWB$ as point H .

Shown above.

- b. Draw the inscribed circle for $\triangle KWB$ using point H , the incenter, as the center of the circle.

Shown above.

- c. Noah is concerned that when he draws the inscribed circle, he does not have the exact points on the triangle where the circle intersects the triangle. He drew three perpendicular lines from the center to each side.



When Circle H is drawn on Noah's version of $\triangle KWB$, what relationship do $\overline{HR}$, $\overline{HT}$, and $\overline{HC}$ have to each other and to Circle H ?

They are the same length because each segment represents a radius of the circle.

- d. Draw Circle H that passes through points T , C , and R . *Students will draw it on the triangle above.*



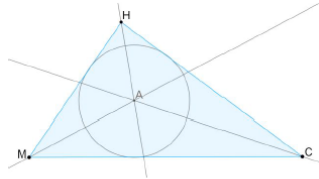
Consider creating, discussing, and displaying a Frayer Model to help students distinguish between the terms "incenter" and "circumcenter."



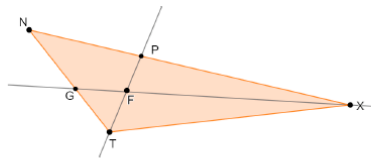
Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the location of the incenter in right, acute, and obtuse triangles.

11.2.2 Exploration: Constructing an Inscribed Circle (continued)

5. Copy $\triangle MHC$ onto another piece of paper. Then, use a compass and a straight edge to inscribe a circle in $\triangle MHC$.



6. $\triangle NXT$ is shown where $\overline{PT}$ bisects $\angle NTX$ and $\overline{GX}$ bisects $\angle PXT$.



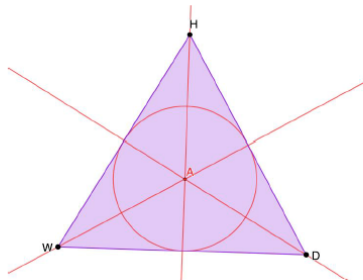
- Will the inscribed circle of $\triangle NXT$ pass through points
- P and G ?

No

- Justify why $\overline{PF}$ and $\overline{FG}$ are not radii of the inscribed Circle F.

The distance from the center of the circle to the sides of the triangle is the perpendicular distance. $\overline{PF}$ and $\overline{FG}$ are not perpendicular to $\overline{NX}$ and $\overline{NT}$, respectively.

7. Copy $\triangle HDW$ onto a piece of patty paper. Then, construct the incenter and circumcenter of $\triangle HDW$.



This is a long and challenging activity. Foster perseverance in students by modeling the steps needed to perform the constructions and recognizing students' efforts during the activity. Develop students' abilities to analyze and solve the problems by using the think aloud strategy to demonstrate expectations for student responses to the problems.



Consider using question 5 as a formative assessment of student mastery of the lesson learning targets. Consider working one-on-one or in a small group with students who are struggling.



- Is it necessary to draw all three angle bisectors to locate the incenter? Why or why not?*
- Is it necessary to construct perpendicular lines from the incenter to each side of the triangle? Why or why not?*



Prompt students to review their work from Lesson 1 to help them construct the circumcenter. Consider having students use different colored writing utensils to construct the incenter than they use to construct the circumcenter. Highlight students' responses as they notice that the incenter and circumcenter are the same point in equilateral triangles.

11.2.3 Wrap-Up: Reflection

Component Overview: Students reflect on the part of the lesson that they found the most difficult or confusing.



11.2.3 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on constructing a circle inscribed in a triangle?

Answers will vary.



This information provided from the students here, coupled with what was observed during the exploration, can provide information for differentiating instruction in later lessons.

11.H1 Inscribing Polygons in Circles

Lesson Introduction

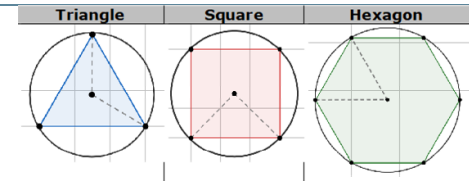
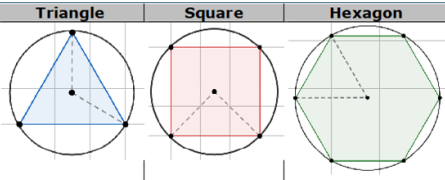
 Benchmark Focus	MA.912.GR.5.4 Construct a regular polygon inscribed in a circle. Regular polygons are limited to triangles, quadrilaterals, and hexagons.		 Learning Targets	- I can construct an equilateral triangle inscribed in a circle. - I can construct a square inscribed in a circle. - I can construct a regular hexagon inscribed in a circle.
 Benchmark Coherence	Building On		Working Toward	
	MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.		N/A	
 Essential Questions	- How do you construct an equilateral triangle inscribed in a circle? - How do you construct a square inscribed in a circle? - How do you construct a regular hexagon in a circle?			
 Lesson Narrative	In this lesson, students discover two methods for performing each of the following constructions: an equilateral triangle inscribed in a circle, a square inscribed in a circle, and a regular hexagon in a circle. Students sort cards to demonstrate their understanding of the sequence of steps needed to complete each construction using each method.			
 Component Summary	11.H1.1 Warm-Up (~5 min.)	Investigate solar system (<i>SE p. 187</i>)	11.H1.3 Activity (20 min.)	Sort cards with construction steps (<i>SE p. 189</i>)
	11.H1.2 Activity (15-20 min.)	Make and test conjectures on how to construct a regular polygon inscribed in a circle (<i>SE p. 188</i>)	11.H1.4 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 191</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Compass - Protractor - Straightedge - Larger Copies of Polygons in 11.H1.2 Activity - Cards for Card Sort (Optional) GeoGebra	
			This is an Honors lesson. Students in standard Geometry should not complete this lesson. Consider moving this activity until after the end of unit assessment if pacing is a concern.	

 Teacher Tips	Common Misconceptions - Students may not understand the difference between the steps in methods 1 and 2 for each construction. - Students may not understand the difference between the steps needed to inscribe an equilateral triangle, a square, and a regular hexagon in a circle.	Things to Consider - In 11.H1.2 Activity, consider modeling the constructions and then having students practice independently a few times as they develop fluency with skills. - Consider creating, discussing, and displaying a Frayer Model to help students distinguish between the steps for each method for each construction. - Consider using a dynamic tool, such as GeoGebra, to provide a visual of each construction. - Triangle: https://www.geogebra.org/m/fbv5s3ed - Square: https://www.geogebra.org/m/Wg5aUDBk - Hexagon: https://www.geogebra.org/m/BgC68JBF
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share During 11.H1.2 Activity, prompt students to initially form conjectures independently. Then, ask students to discuss their conjectures with a partner. Finally, have a variety of students share their conjectures with the whole class during the debrief discussion. After each conjecture is shared, ask a variety of students if they agree or disagree and why.	MA.K12.MTR.5.1: Patterns and Structure During 11.H1.2 and 11.H1.3 Activities, students explore different methods of constructing regular polygons inscribed in circles. Pose purposeful questions and engage students in discussions to ensure that they understand the sequences of steps for each method of each construction.
 Differentiate Instruction	Discussions during 11.H1.2 and 11.H1.3 Activities provide an auditory entry point for students to form conjectures about how to construct each regular polygon inscribed in a circle. The diagrams in 11.H1.2 Activity provide a visual entry point for students to see each regular polygon inscribed in a circle. Provide a kinesthetic entry point by prompting students to use a compass and straightedge to check the sequences of steps they decide on in 11.H1.3 Activity by seeing if the steps result in the correct constructions.	
Lesson Notes:		

Triangle

Square

Hexagon



11.H1.1 Warm-Up: Earth's Orbit

Component Overview: Students are engaged with two interesting, real-world problems related to the diameter of the sun and the distance to the sun.



11.H1.1 Warm-Up: Earth's Orbit



Since Earth's orbit is an ellipse, the Earth is slightly closer to the sun in January. The picture shows two photos taken by the SOHO satellite. The left side shows the sun on January 4, 2009 and the right side shows the sun on July 4, 2009.



Consider inviting an Astronomer or Physicist to join the class, virtually or in-person. Ask the guest speaker how they use geometry to help students see the utility of the topics they are studying. Also, prompt students to prepare questions for the guest speaker.

A scientist measured the two images. She found that the diameter of the image for January was 72 millimeters and the image for July was 69 millimeters.

1. By what percentage did the diameter of the Sun change between January and July compared to its average diameter of January and July?
2.1%
2. If the average distance to the Sun from Earth is 149,600,000 kilometers, how much closer is Earth to the Sun in January compared to July?
6,283,200 kilometers

11.H1.2 Activity: Regular Polygons Inscribed in Circles

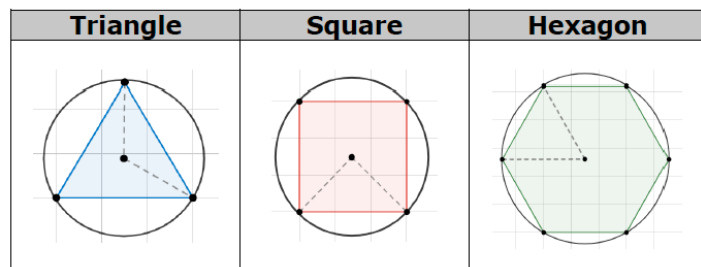
Component Overview: Students make and test conjectures as they discover how to construct regular polygons inscribed in circles. Students will each need a copy of the polygons, compasses, and straightedges.



11.H1.2 Activity: Regular Polygons Inscribed in Circles

Work with your partner to answer each question. Use the larger copies of each polygon provided by your teacher.

1. Use a compass to draw a circumscribed circle for each polygon.
2. Mark the center of the circle that was drawn on the polygon. Draw a central angle that passes through two subsequent vertices, as shown.



3. Use a protractor to determine the measure of each central angle.

	Triangle	Square	Hexagon
Measure of Central Angle	120°	90°	60°



Students may make errors during this activity when they are writing a conjecture about how to construct each regular polygon inscribed in a circle. Allow the opportunity for students to correct their own thinking as they work through with their partner. Do not be afraid to let the mistakes happen, only to discuss them during the transition to the next activity.



- What do you notice about the measure of the central angle of each of the shapes?
- How does the measure of the central angle relate to the number of sides of the polygon?

11.H1.2 Activity: Regular Polygons Inscribed in Circles (continued)

4. Make a conjecture about how to construct each regular polygon inscribed in a circle using a compass and a straight edge.

	Construction
Triangle	Answers will vary.
Square	Answers will vary.
Hexagon	Answers will vary.

5. Ask another pair of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first three columns of the table.* Have the person initial next to your summary stating that your summary was accurate.

	Other Partner Pair's Conjecture	Other Partner Pair's Initials
Triangle	Answers will vary.	
Square	Answers will vary.	
Hexagon	Answers will vary.	

6. Review your conjectures with your partner and if necessary, make any changes.
Answers will vary.



Think-Pair-Share

Prompt students to initially form conjectures independently. Then, ask students to discuss their conjectures with a partner. Finally, have a variety of students share their conjectures with the whole class during the debrief discussion. After each conjecture is shared, ask a variety of students if they agree or disagree and why.

11.H1.3 Activity: Card Sort

Component Overview: Students sort cards to demonstrate their understanding of the sequence of steps needed to construct an equilateral triangle, a square, and a regular hexagon inscribed in a circle, using two different methods for each construction.



11.H1.3 Activity: Card Sort

Sort the sets for each construction. Then, write the steps in the table.

	Construction Steps
Triangle Method 1	1. Use a compass to draw a circle. Mark the center and label it. 2. Keep the span of the compass used to draw the circle. 3. Mark any point on the circumference of the circle. 4. Place the endpoint of the compass on the marked point and make a small arc on the circumference of the circle. 5. Continue around the circle making a total of 5 small arcs. 6. Place a point at every other arc. 7. Connect the points
Square Method 1	1. Use a compass to draw a circle. Mark the center and label it. 2. Draw a diameter. Label the endpoints. 3. Construct a perpendicular bisector of the diameter. Label the points that intersect with the circle. 4. Connect the points.



For this activity, consider allowing students to work with a partner or small group, and utilize the instructional routine “Take Turns.” Each group member can take a turn at determining the next step of each construction. Students can agree or disagree with a justification and ask clarifying questions during this time.



Provide a kinesthetic entry point by prompting students to use a compass and straightedge to check the sequences of steps they decide on by seeing if the steps result in the correct constructions. Discuss the sequences of steps to provide an auditory entry point.

11.H1.3 Activity: Card Sort (continued)

Hexagon Method 1	1. Use a compass to draw a circle. Mark the center and label it. 2. Keep the span of the compass used to draw the circle. 3. Mark any point on the circumference of the circle. 4. Place the endpoint of the compass on the marked point and make a small arc on the circumference of the circle. 5. Continue around the circle making a total of 5 small arcs. 6. Place a point at every arc. 7. Connect the points.
Triangle Method 2	1. Use a compass to draw a circle. Mark the center and label it. 2. Keep the span of the compass used to draw the circle. 3. Draw a diameter. Label the endpoints. 4. Place the endpoint of the compass on one of the endpoints of the diameter. 5. Draw a circle. 6. Label the intersections of the new circle with the original circle. 7. Connect the intersections with the other endpoint of the diameter.



What are the similarities and differences between methods 1 and 2 for constructing a triangle, a square, and a hexagon inscribed in a circle?

11.H1.3 Activity: Card Sort (continued)

Square Method 2

1. Use a compass to draw a circle. Mark the center and label it.
2. Draw a diameter. Label the endpoints.
3. Construct a circle with a radius larger than the original circle with center at one endpoint of the diameter.
4. Construct a circle with a radius larger than the original circle with center at the other endpoint of the diameter.
5. Mark the intersections of the two circles.
6. Draw a line through the intersection of the two circles and the original circle's center.
7. Mark the two intersections of the line and the original circle. Label the points.
8. Connect the points.



Consider creating, discussing, and displaying a Frayer Model to help students distinguish between the steps for each method for each construction.

11.H1.3 Activity: Card Sort (continued)

Hexagon Method 2	1. Use a compass to draw a circle. Mark the center and label it. 2. Keep the span of the compass used to draw the circle. 3. Draw a diameter. Label the endpoints. 4. Place the endpoint of the compass on one of the endpoints of the diameter. 5. Draw a circle. 6. Label the intersections of the new circle with the original circle. 7. Place the endpoint of the compass on the other endpoint of the diameter. 8. Draw a circle. 9. Label the intersections of the new circle with the original circle. 10. Connect the intersections of both circles and the endpoints of the diameter.
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Consider using a dynamic tool, such as GeoGebra, to provide a visual of each construction.

11.H1.4 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



11.H1.4 Wrap-Up: Cool Down

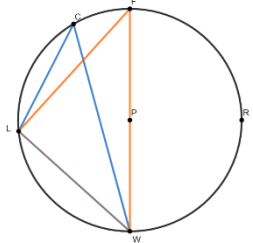
1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about inscribing polygons in circles.

Answers will vary.

11.3 Solving Problems Involving Triangles Inscribed in Circles

Lesson Introduction

 Benchmark Focus	MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.		 Learning Targets	I can solve mathematical problems involving triangles inscribed in a circle.
 Benchmark Coherence	Building On			Working Toward
	N/A			N/A
 Essential Questions	- How do you solve problems involving triangles inscribed in a circle? - What is the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles?			
 Lesson Narrative	In this lesson, students will apply their knowledge of triangles inscribed in a circle to solve mathematical problems. Students initially explore the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles. Next, students practice solving problems requiring them to use the relationship to find angle and arc measurements.			
 Component Summary	11.3.1 Warm-Up (~5 min.)	Solve a number puzzle (<i>SE p. 192</i>)	11.3.3 Guided Practice (10-15 min.)	Guided Practice solving problems using the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles (<i>SE p. 193</i>)
	11.3.2 Exploration (15 min.)	Explore the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles (<i>SE p. 192</i>)	11.3.4 Your Turn! (10 min.)	Independent practice solving problems using the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles (<i>SE p. 195</i>)
	11.3.5 Wrap-Up (~5 Min)	Students summarize their learning (<i>SE p. 196</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Inscribed Angle - Intercepted Arc		- Rulers - Patty Paper (Optional) Computers	
			Additional Preparation	
			Prior to the lesson, work through 11.3.2 Exploration, 11.3.3 Guided Practice, and 11.3.4 Your Turn! to ensure that you are familiar with the steps students will take and are prepared to support them as they work.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not understand the relationship between the inscribed angle and the intercepted arc. - Students may not understand that inscribed angles in different triangles can have the same measurement if they share the same intercepted arc. - Students may not understand that the angle or arc measurements are represented by evaluated algebraic expression, and may provide the value of the variable.	- Consider creating an anchor chart with the relationship between the inscribed angle and the intercepted arc during the debrief of 11.3.2 Exploration. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the relationship between the inscribed angle and in the intercepted arc.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Take Turns In 11.3.3 Guided Practice, students have an opportunity to solve problems using the relationship between the inscribed angle and intercepted arc in triangles inscribed in circles. Prompt students to work with a partner and take turns explaining how they solved the problem and sharing if they agree or disagree and why.	MA.K12.MTR.3.1: Mathematical Fluency Students first explore the relationship between the inscribed angle and the intercepted arc in triangles inscribed in circles during 11.3.2 Exploration. Next, they practice solving a variety of problems, first guided in 11.3.3 Guided Practice, and then independently in 11.3.4 Your Turn!. This scaffolded practice will help students build knowledge, skills, and confidence with the lesson learning target.
	This lesson provides multiple entry points for visual learners by providing diagrams to represent each question. Auditory learners will have opportunities to discuss with peers the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles. The constructions in 11.3.2 Exploration provides a hands-on approach for kinesthetic learners.	
Lesson Notes:		

11.3.1 Warm-Up: Patterns

Component Overview: Students investigate a number pattern displayed on rocks arranged in a triangle. Students first pose questions about the pattern and then describe the pattern. Solving non-routine problems, such as this puzzle, can help students build strategic competence.



11.3.1 Warm-Up: Patterns



Notice and Wonder

Prompt students to reflect on what they notice and wonder about the pattern. Call on a variety of students to share what they noticed and wondered. This activity provides a low floor, as all students can participate, and high ceiling, as students can be challenged to think of multiple noticings and wonderings.

1. What is the first question that comes to your mind?
Answers will vary. What happened to the numbers in the middle? Where is the 5?
2. Describe the pattern used in the image.
The left-hand side of the rocks is being multiplied by two each time. The right-hand side of the rocks is being multiplied by three each time.

11.3.2 Exploration: Triangles Inscribed in Circles

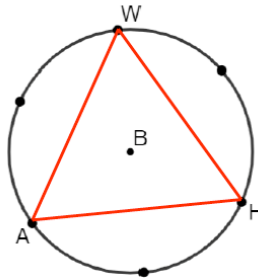
Component Overview: Students explore the relationship between inscribed angles and intercepted arcs in an equilateral triangle inscribed in a circle.



11.3.2 Exploration: Triangles Inscribed in Circles

Work with your partner to complete the following.

Elijah is constructing a triangle inscribed in Circle B and stopped after he created six points on Circle B that are the same distance from each other.



- Using a straightedge, draw a line from point A to point H .

Answer on image above

- Using a straightedge, draw a line from point H to point W .

Answer on image above

- In relation to the circle, what type of angle is $\angle AHW$?

An inscribed angle

- Explain the relationship between $m\angle AHW$ and $m\widehat{WA}$.

$m\angle AHW$ is $\frac{1}{2}(m\widehat{WA})$ because an inscribed angle is half the measure of its intercepted arc.



This Exploration is designed to allow students to engage with it with a partner, rather than being guided through by the teacher. Observe and listen as students work with their partners. There will be an opportunity to summarize and synthesize what is discovered in the next component.



Consider creating an anchor chart with the relationship between the inscribed angle and the intercepted arc.



Encourage students to discuss their explanation for question 4 with their partner to provide an auditory entry point. The use of a ruler, straightedge, and/or patty paper provide an entry point for kinesthetic learners.

11.3.2 Exploration: Triangles Inscribed in Circles (continued)

5. Using a straightedge, draw a line from point A to point W .

Answer on image above

6. Use a ruler or patty paper to verify that $\triangle AHW$ is an equilateral triangle.

Answers will vary.

7. What is the $m\angle AHW$?

60°

8. What is the $m\widehat{WA}$?

120°

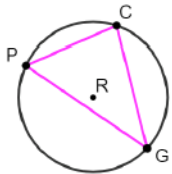
11.3.3 Guided Practice: Triangles Inscribed in Circles

Component Overview: Students engage in Guided Practice by solving problems using the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles.

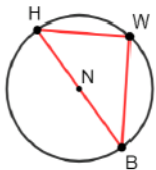


11.3.3 Guided Practice: Triangles Inscribed in Circles

1. Circle R is shown. Points P , C , and G are on circle R and create $\triangle PCG$. $m\widehat{PG} = 162^\circ$, $m\angle CPG = 58^\circ$, and $m\angle PCG = (7x - 17)^\circ$.



- What is the $m\angle PCG$?
 81°
 - Find the value of x .
 $x = 14$
 - What is the $m\angle PGC$?
 41°
2. Circle N has points H , W , and B on the circle and diameter $\overline{HB}$. $\triangle HWB$ is inscribed in circle N . $m\widehat{BW} = 102^\circ$ and $m\angle HBW = (4x - 10)^\circ$.



- What is the $m\angle HBW$?
 39°
- Find the value of x .
 $x = 12.25$



Use observational data from 11.3.2 Exploration to determine how much guidance is needed to bring together key ideas in this Guided Practice. If necessary, consider modeling question 1 or pulling a small group as they work through that problem to address common misconceptions.

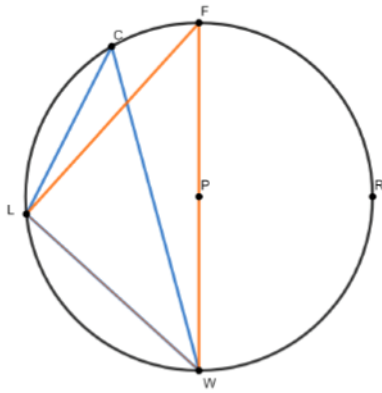


Take Turns

Students have an opportunity to solve problems using the relationship between the inscribed angle and intercepted arc in triangles inscribed in circles. Prompt students to work with a partner and take turns explaining how they solved the problem and sharing if they agree or disagree and why.

11.3.3 Guided Practice: Triangles Inscribed in Circles (continued)

3. $\triangle CLW$ and $\triangle FLW$ are inscribed in Circle P , where $\overline{FW}$ is the diameter of Circle P . $m\widehat{FC} = (2x - 6)^\circ$, $m\widehat{LC} = (4x - 6)^\circ$, and $m\widehat{LW} = (3x + 30)^\circ$.



Complete the table.

Angle Measures	Angle Measures
$m\angle LCW = 42^\circ$	$m\angle LFW = 42^\circ$
$m\angle CWL = 33^\circ$	$m\angle FWC = 15^\circ$
$m\angle FLC = 15^\circ$	$m\angle FLW = 90^\circ$



Consider prompting students to separate this circle into two separate circles, with one triangle inscribed on each circle. Students can trace the circle and inscribed triangles on patty paper.

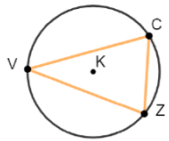
11.3.4 Your Turn!

Component Overview: Students practice solving problems independently using the relationship between inscribed angles and intercepted arcs in triangles inscribed in circles. Circulate while students work, to support struggling students and determine which students you will ask to share their strategies during a debrief discussion.



11.3.4 Your Turn!

1. Circle K has points V , C , and Z on the circle, and $\triangle VCZ$ is inscribed in circle K . $m\widehat{VC} = (9x + 23)^\circ$, $m\angle CVZ = 36^\circ$, and $m\angle VZC = (7x - 21)^\circ$.



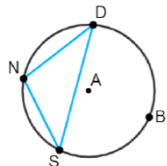
- a. What is the value of x ?

$$x = 13$$

- b. What is the $m\angle VCZ$?

$$74^\circ$$

2. Circle A has points N , D , B , and S on the circle, and $\triangle SDN$ is inscribed in Circle A . $m\widehat{ND} = (7x - 7)^\circ$, $m\widehat{DBS} = (23y - 5)^\circ$, $m\angle NSD = (5x - 23)^\circ$, and $m\angle DNS = (8y + 29)^\circ$.



Complete the table.

Angle Measures	Arc Measures
$m\angle NSD = 42^\circ$	$m\widehat{ND} = 84^\circ$
$m\angle SND = 101^\circ$	$m\widehat{DBS} = 202^\circ$
$m\angle NDS = 37^\circ$	$m\widehat{NS} = 74$



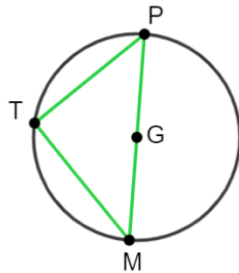
Students first explore the relationship between the inscribed angle and the intercepted arc in triangles inscribed in circles during 11.3.2 Exploration. Next, they practice solving a variety of problems, first guided in 11.3.3 Guided Practice, and then independently in 11.3.4 Your Turn!. This scaffolded practice will help students build knowledge, skills, and confidence with the lesson learning target. Debrief each component with a discussion to check for student understanding.



Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the relationship between the inscribed angle and in the intercepted arc.

11.3.4 Your Turn! (continued)

3. Circle G has diameter $\overline{MP}$ and $\triangle MTP$ inscribed in the circle. $m\angle TMP = (6x - 8)^\circ$ and $m\angle TPM = (4x + 13)^\circ$.



- Find the value of x .
 $x = 8.5$
- What is the $m\angle TMP$?
 43°
- What is the $m\angle TPM$?
 47°



- How did you determine the inscribed angle measures and the intercepted arc measure?
- Why is it important that the problem stated that the diameter of circle G was MP ?
- How could you verify that your angle measures are correct?

11.3.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



11.3.5 Wrap-Up: Cool Down

1. Write a summary of what you learned about solving problems involving triangles inscribed in circles.

Answers will vary.



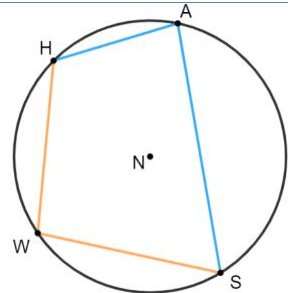
Consider providing students with sentence stems or answer options to choose from. These strategies will help students compose written responses to the prompt.

11.4 Solving Problems Involving Quadrilaterals Inscribed in Circles

Lesson Introduction

 Benchmark Focus	MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.		 Learning Targets	I can solve mathematical problems involving quadrilaterals inscribed in a circle.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.		N/A	
 Essential Questions	- How do you solve mathematical problems involving quadrilaterals inscribed in a circle? - What is the relationship between the opposite angles in quadrilaterals inscribed in circles?			
 Lesson Narrative	In this lesson, students decompose quadrilaterals inscribed in circles into two triangles. They use the relationship between inscribed angles and intercepted arcs to find unknown angles and arcs. Through guided and independent practice, students discover that the opposite angles in quadrilaterals inscribed in circles are supplementary.			
 Component Summary	11.4.1 Warm-Up (~5 min.)	Notice and Wonder about Machu Picchu (<i>SE p. 197</i>)	11.4.3 Guided Practice (10-15 min.)	Guided Practice finding angles in quadrilaterals inscribed in circles (<i>SE p. 199</i>)
	11.4.2 Exploration (15 min.)	Make conjectures about angles in quadrilaterals inscribed in circles (<i>SE p. 198</i>)	11.4.4 Your Turn! (10 min.)	Independent practice finding angles in quadrilaterals inscribed in circles (<i>SE p. 200</i>)
	11.4.5 Wrap-Up (~5 Min)	Identify and correct errors in sample student work (<i>SE p. 201</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Inscribed Angle - Intercepted Arc - Supplementary		(Optional) Colored Pencils	Prior to the lesson, work through 11.4.2 Exploration, 11.4.3 Guided Practice, and 11.4.4 Your Turn! to ensure that you are familiar with the steps students will take and are prepared to support them as they work.

 Teacher Tips	Common Misconceptions - Students may not understand the relationship between the inscribed angle and the intercepted arc. - Students may struggle with identifying opposite angles. - Students may solve for opposite angles using congruency instead of supplementary.	Things to Consider - Consider displaying the anchor chart from Lesson 3 with the relationship between the inscribed angle and the intercepted arc for students to reference. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the relationship between the opposite angles in quadrilaterals inscribed in circles.
	Literacy and Language Routines Notice and Wonder During 11.4.1 Warm Up, prompt students to reflect on what they notice and wonder about Machu Picchu. Call on a variety of students to share what they noticed and wondered. This activity provides a low floor, as all students can participate, and high ceiling, as students can be challenged to think of multiple noticings and wonderings.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection In 11.4.5 Wrap-Up, students have the opportunity to demonstrate their mastery of the learning target by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors themselves in the future.
 Differentiate Instruction	Provide an auditory entry point in 11.4.2 Exploration by having students discuss their conjectures about the relationships of the angles in quadrilaterals inscribed in circles. Provide a visual entry point by prompting students to color code the opposite angles in the diagrams in 11.4.3 Guided Practice. Provide a kinesthetic entry point in 11.4.4 Your Turn! by having students construct their own diagram and create problems involving a quadrilateral inscribed in a circle.	
Lesson Notes:		



11.4.1 Warm-Up: Notice and Wonder

Component Overview: Students are engaged by a photograph and written description of Machu Picchu. Then, they are asked to share what they notice and wonder about Machu Picchu. This component exposes students to a historic real-world application, helping them see the utility of geometry.



11.4.1 Warm-Up: Notice and Wonder

Machu Picchu is an Inca citadel in southern Peru. Machu Picchu was built in the classical Inca style with polished dry-stone walls and consists of more than 150 buildings. Although many of the stones that were used to build the city were more than 50 pounds, it is believed that no wheels were used to transport these rocks up the mountain.



This activity provides students with an image of Machu Picchu, Peru. Students are asked to notate what they notice from the image and what they wonder from the image. This routine provides a low entry point for students to engage in general discussion.

1. What do you notice?

Answers will vary. I noticed that there are a lot of houses, which means a lot of carrying of heavy stones. I noticed that there is an open space in between buildings.

2. What do you wonder?

Answers will vary. I wonder how many people they had to carry the stones needed. I wonder if the materials were stored in the open green space.

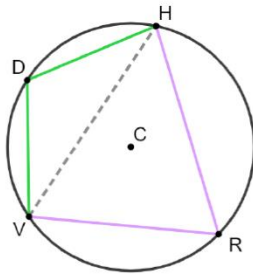
11.4.2 Exploration: Quadrilaterals Inscribed in Circles

Component Overview: Students make, discuss, and revise conjectures about angles in quadrilaterals inscribed in circles. Encourage students to support their conjectures with justifications and ask others if they agree or disagree and why.



11.4.2 Exploration: Quadrilaterals Inscribed in Circles

1. Circle C is shown, where $\overline{DV} \cong \overline{DH}$. $m\widehat{RV} = 100^\circ$, $m\widehat{DV} = 68^\circ$, $m\widehat{HDV} = 136^\circ$, and $m\widehat{HRV} = 224^\circ$.



- a. Complete the table.

Angle Measures	Angle Measures
$m\angle RHV = 50^\circ$	$m\angle HDV = 112^\circ$
$m\angle DHV = 34^\circ$	$m\angle DVH = 34^\circ$
$m\angle HRV = 68^\circ$	$m\angle HVR = 62^\circ$

- b. Work with your partner to make a conjecture about the relationships of the angles in $DHRV$.
Answers will vary. The opposite angles are supplementary.
- c. Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials
<i>Answers will vary</i>		
<i>Answers will vary</i>		



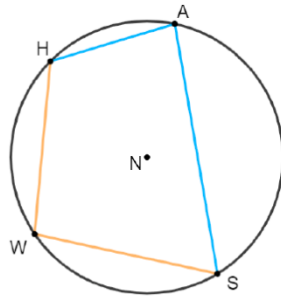
For this explanation, allow students to work with a partner to answer the scaffolded questions, rather than being guided through by a teacher. There will be an opportunity in the next component to summarize and synthesize what is discovered.



Consider displaying the anchor chart from Lesson 3 with the relationship between the inscribed angle and the intercepted arc for students to reference.

11.4.2 Exploration: Quadrilaterals Inscribed in Circles (continued)

- d. Review your conjecture with your partner. If necessary, make any changes.
2. Circle N has points H , A , S , and W on the circle. Quadrilateral $HASW$ is inscribed in the circle. $m\widehat{HA} = 59^\circ$, $m\widehat{HWS} = 166^\circ$, and $m\widehat{WHA} = 136^\circ$.



Use your conjecture to find the measures.

Angle Measures

$$\begin{aligned} m\angle WHA &= 112^\circ \\ m\angle HAS &= 83^\circ \\ m\angle ASW &= 68^\circ \\ m\angle SWH &= 97^\circ \end{aligned}$$

Arc Measures

$$\begin{aligned} m\widehat{SA} &= 135^\circ \\ m\widehat{SW} &= 89^\circ \\ m\widehat{HW} &= 77^\circ \end{aligned}$$



Some students may benefit from sentence stems or answer choices to help them form their conjectures.



Collect and Display

As students share their conjectures, collect a variety of their conjectures, and display them for all to see. Then, engage students in a discussion about whether they agree or disagree and why.

11.4.3 Guided Practice: Quadrilaterals Inscribed in Circles

Component Overview: Students engage in Guided Practice finding angles in quadrilaterals inscribed in circles.



11.4.3 Guided Practice: Quadrilaterals Inscribed in Circles

1. Complete the statement.

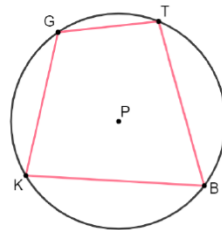
If a quadrilateral is inscribed in a circle,

- ☐ adjacent
☐ opposite

angles are

- ☐ congruent.
☐ complementary.
☐ supplementary.

2. Circle P has $GTBK$ inscribed in the circle. $m\widehat{TBK} = 206^\circ$ and $m\angle GKB = 84^\circ$.



- a. What is the $m\angle TBK$?
 77°
- b. What is the $m\angle GTB$?
 96°



Check for understanding after question 1 to make sure that students understand the relationship between the angles in quadrilaterals inscribed in circles. Use this data to determine how much guidance is needed in this Guided Practice component.



- What does it mean for two angles to be supplementary?
- Which angles in the quadrilateral are opposite angles?

11.4.3 Guided Practice: Quadrilaterals Inscribed in Circles (continued)

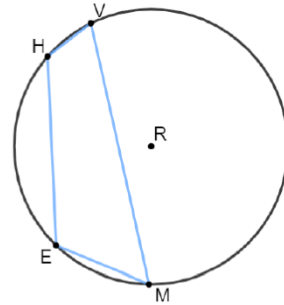
3. Circle R has $VMEH$ inscribed in the circle.

$$m\angle HEM = (2x - 7)^\circ,$$

$$m\angle EHV = (2x + 3)^\circ, \text{ and}$$

$$m\angle HVM = (x + 4)^\circ.$$

- a. Find the value of x .
 $x = 61$



Provide a visual entry point by prompting students to color code the opposite angles in the diagrams.

- b. Complete the table.

$m\angle HVM = 65^\circ$
$m\angle HEM = 115^\circ$
$m\angle EHV = 125^\circ$
$m\angle EMV = 55^\circ$

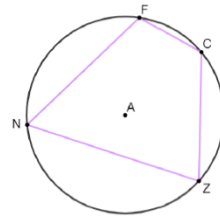
11.4.4 Your Turn!

Component Overview: Students engage in independent practice finding angles in quadrilaterals inscribed in circles. Circulate while students work, to support struggling students and determine which students to call on during the debrief discussion.



11.4.4 Your Turn!

1. Circle A has $NFCZ$ inscribed in the circle. $m\angle NFC = (6x + 32)^\circ$, $m\angle FCZ = (6y - 8)^\circ$, $m\angle NZC = 73^\circ$, and $m\angle FNZ = (3y - 1)^\circ$.



Challenge students who finish early by asking them to construct their own diagram and create problems involving a quadrilateral inscribed in a circle. Students can swap problems with a partner and solve each other's problems.

- a. Find the value of x .

$$x = 12.5$$

- b. What is the $m\angle NFC$?

$$107^\circ$$

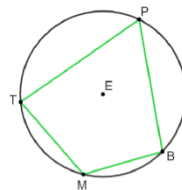
- c. Find the value of y .

$$y = 21$$

- d. What is the $m\angle FCZ$?

$$118^\circ$$

2. Circle E has $TPBM$ inscribed in the circle. $m\widehat{TM} = 71^\circ$, $m\widehat{MB} = 61^\circ$, and $m\angle TMB = (12x - 3)^\circ$.



- a. Find the value of x .

$$x = 9.75$$

- b. What is the $m\angle TMB$?

$$114^\circ$$

11.4.5 Wrap-Up: Error Analysis

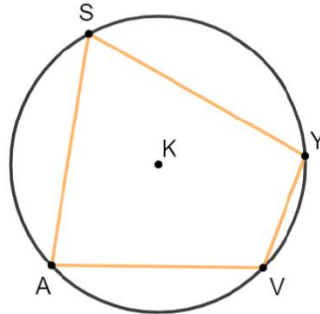
Component Overview: Students engage in analyzing a sample of student work for errors.



11.4.5 Wrap-Up: Error Analysis

1. $SYVA$ is inscribed in Circle K , where
 $m\angle ASY = (3x + 23)^\circ$ and
 $m\angle AVY = (6x + 13)^\circ$.

Anastasia is finding the value of x . Her work is shown.



Students have the opportunity to demonstrate their mastery of the learning target by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors in the future.

Anastasia's Work

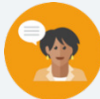
$$\begin{aligned} 3x + 23 &= 6x + 13 \\ 3x &= 10 \\ x &= \frac{10}{3} \end{aligned}$$

- Write a note to Anastasia explaining her mistake.
Answers will vary. Anastasia, opposite angles of an inscribed quadrilateral are supplementary. To solve for x , add the opposite angles and set them equal to 180° .
- Find the value of x .
 $x = 16$

11.5 Interior and Exterior Angles of Triangles

Lesson Introduction

 Benchmark Focus	MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.		 Learning Targets	- I can show that the sum of the interior angles of a triangle is 180°. - I can show that the sum of the exterior angles of a triangle is 360°. - I can solve problems involving postulates, relationships, and theorems of triangles.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.		N/A	
 Essential Questions	- How can you show that the sum of interior angles of a triangle is 180°? - How can you show that the sum of the exterior angles of a triangle is 360°?			
 Lesson Narrative	Students discover how to show that the sum of the interior angles of a triangle is 180° , and that the sum of the exterior angles of a triangle is 360° . Students make, test, and revise conjectures based on relationships among angles that they observe in geometric constructions, including a circumscribed circle of a triangle, a circle inscribed in a triangle, and the sectors formed from the exterior angles of a triangle.			
 Component Summary	11.5.1 Warm-Up (~5 min.)	Share and justify superpowers of geometry superhero and villain (<i>SE p. 202</i>)	11.5.3 Activity (10-15 min.)	Discover relationships among exterior angles of triangles (<i>SE p. 204</i>)
	11.5.2 Exploration (15 min.)	Make conjectures about the relationship among interior angles of triangles (<i>SE p. 203</i>)	11.5.4 Guided Practice (10 min.)	Practice finding interior and exterior angles of triangles (<i>SE p. 206</i>)
	11.5.5 Wrap-Up (~5 Min)	Students summarize their learning (<i>SE p. 207</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Exterior Angle - Triangle Sum Theorem <u>Additional Glossary:</u> - Interior Angle		- Paper - Scissors - Straightedge - Compass (Optional) Colored Pencils	
	Additional Preparation Prior to the lesson, work through 11.5.2 Exploration, 11.5.3 Activity, and 11.5.4 Guided Practice to ensure that you are familiar with the steps students will take and be prepared to support them as they work.			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not understand that the sum of the interior angles of a triangle is 180°. - Students may not understand that the sum of the exterior angles of a triangle is 360°. - Students may understand the relationships but make errors in their calculations.	- Consider creating and displaying an Anchor Chart with the relationship between the interior and exterior angles in triangles for students to reference. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the relationship between the interior and exterior angles in triangles.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 11.5.2 Exploration, prompt students to write down things they notice and wonder about the interior angles of the triangles from question 1 and question 2. This will allow students to talk through the relationship of the interior angles of triangles before writing a conjecture.	MA.K12.MTR.4.1: Discussion and Reflection During 11.5.2 Exploration and 11.5.3 Activity, students write a conjecture about the relationship among interior and exterior angles of a triangle. Students will have the opportunity to discuss their conjectures with their classmates. Encourage students to reflect on their initial conjectures and revise them as needed based on the conversations they have with their peers.
 Differentiate Instruction	Consider creating an anchor chart that displays all the relationships discovered throughout this unit, along with a specific example or two of each being applied as a reference point for students to use as they complete this lesson and the remainder of the unit.	
Lesson Notes:		

11.5.1 Warm-Up: Superhero

Component Overview: Students share and justify the superpowers of a geometry superhero and villain.



11.5.1 Warm-Up: Superhero

1. If geometry had a superpower, it would be ...

Answers will vary

because ...

Answers will vary

2. Geometry's villain would be ...

Answers will vary

because ...

Answers will vary



These prompts are intended to inspire student curiosity and interest. This component can help build students' agency and mathematical identities, as each student can participate. Consider having students draw and label a diagram of the superhero and villain.

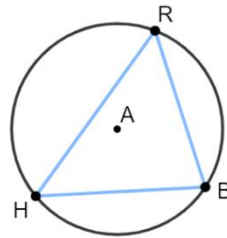
11.5.2 Exploration: Interior Angles of Triangles

Component Overview: Students discover how to show that the sum of the interior angles of a triangle is 180° . Students make, test, and revise conjectures based on relationships among angles that they observe in a circumscribed circle of a triangle and a circle inscribed in a triangle.



11.5.2 Exploration: Interior Angles of Triangles

1. Circle A is shown, where $m\widehat{RB} = 102^\circ$, $m\widehat{BH} = 108^\circ$, and $m\widehat{HR} = 150^\circ$.

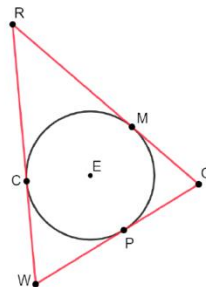


- a. Complete the table.

Angle Measures	
$m\angle RHB =$	51°
$m\angle RBH =$	75°
$m\angle HRB =$	54°

- b. What is the sum of the angles in $\triangle RBH$?
 180°

2. Circle E is shown where points P , C , and M are on the circle. $\overline{RG}$ and $\overline{RW}$ are tangent to Circle E and intersect at point R outside the circle. $\overline{GR}$ and $\overline{GW}$ are tangent to Circle E and intersect at point G outside the circle. $\overline{WR}$ and $\overline{WG}$ are tangent to Circle E and intersect at point W outside the circle. $m\widehat{CM} = 136^\circ$, $m\widehat{MP} = 108^\circ$, and $m\widehat{CP} = 116^\circ$.



- a. Complete the table.

Angle Measures	
$m\angle WRG =$	44°
$m\angle RGW =$	72°
$m\angle GWR =$	64°



This activity is an exploration of interior angles of triangles. Be careful not to give away too much information when supporting students during this activity. This activity has a natural unfolding process through discovery and exploration.

11.5.2 Exploration: Interior Angles of Triangles (continued)

- b. What is the sum of the angles in $\triangle RGW$?
 180°
- c. Write a conjecture about the sum of the interior angles of a triangle.
The sum of the interior angles of a triangle is 180° .
- d. Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials
<i>Answers will vary</i>		

- e. Review your conjecture with your partner. If necessary, make any changes.



Triangle Sum Theorem

The sum of the measures of the interior angles of a triangle is 180° .



Notice and Wonder

Prompt students to write down things they notice and wonder about the sum of the interior angles of the triangles from question 1 and question 2. This will allow students to talk through the relationship of the interior angles of triangles before writing a conjecture.



Allowing students to share their conjecture with peers will provide entry points for auditory learners. Prompt students' by asking them if they agree or disagree with their peers' conjectures and why.

11.5.3 Activity: Exterior Angles of Triangles

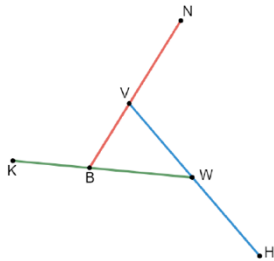
Component Overview: Students discover how to show that the sum of the exterior angles of a triangle is 360° . Students make, test, and revise conjectures based on relationships among angles that they observe in a geometric construction of the sectors formed from the exterior angles of a triangle.



11.5.3 Activity: Exterior Angles of Triangles

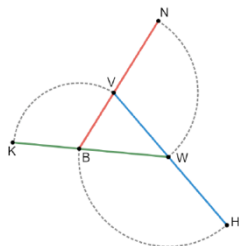
Work with your partner to complete the following.

- On a separate sheet of paper, draw three line segments, $\overline{BN}$, $\overline{VH}$, and $\overline{KW}$, to make $\triangle VWB$ as shown.



An **exterior angle** is an angle created by extending one leg of a triangle.

- List the exterior angles of $\triangle VWB$.
 $\angle VBK$, $\angle NVW$, $\angle BWH$
 - Estimate the sum of the exterior angles of $\triangle VWB$.
Answers will vary. 360°
- Use a compass to draw $\widehat{BH}$ using point W as the center, $\widehat{KV}$ using point B as the center, and $\widehat{NW}$ using point V as the center, as shown.



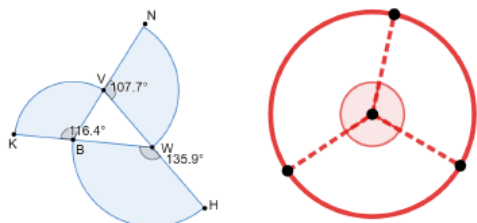
For this activity, provide students with a piece of paper, scissors, and a straightedge. Students will work through the scaffolded questions with a partner to discover that the sum of the exterior angles of a triangle is 360° . Consider reviewing this activity prior to the lesson to assist struggling students.



Consider providing struggling students access to a video that explains the steps of discovering that the sum of the exterior angles of a triangle is 360° . This will allow students the opportunity to pause and rewind the video as needed.

11.5.3 Activity: Exterior Angles of Triangles (continued)

- a. Trace the figure above onto patty paper and cut out each sector. Rearrange the sectors to determine the sum of the exterior angles.



- b. What do you notice about the sum of the exterior angles of the triangle?
They add to be 360° .
- c. Write a conjecture about the sum of the exterior angles of a triangle.
The sum of exterior angles of a triangle is 360° .
- d. Ask two classmates for their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture	My Summary of Their Reason	Initials
<i>Answers will vary</i>		
<i>Answers will vary</i>		

- e. Review your conjecture with your partner. If necessary, make any changes.



Consider using a dynamic tool, such as GeoGebra, to enable visual and kinesthetic students to investigate the relationship between the interior and exterior angles in triangles.



Consider challenging students who finish this activity early by having students verify if their conjecture will always be true. Students can create three line segments that form a triangle and complete the steps in question 2 to confirm if their conjecture is still true.



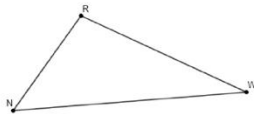
Listen as students discuss their thinking in response to their conjecture to determine student thinking to highlight, and/or misconceptions to address.

11.5.4 Guided Practice: Interior and Exterior Angles of Triangles

Component Overview: Students practice finding interior and exterior angles of triangles. Encourage students to show all of their work and check their initial calculations. During the debrief discussion, call on a variety of students to share their strategies and answers. Remind students to include the degree symbol with their answers.

11.5.4 Guided Practice: Interior and Exterior Angles of Triangles

1. $\triangle RWN$ is shown where $m\angle NRW = 14x + 3$, $m\angle RWN = \frac{9}{2}x - \frac{5}{2}$, and $m\angle WNR = 9x - 13$.

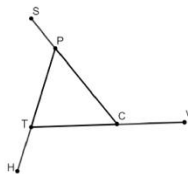


- a. Find the value of x .
 $x = 7$

- b. Complete the table.

$m\angle NRW = 101^\circ$	$m\angle RWN = 29^\circ$	$m\angle WNR = 50^\circ$
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2. $\triangle PCT$ is shown, where $\angle SPT$, $\angle PCV$, and $\angle HTC$ are exterior angles. $m\angle PCT = 52^\circ$, $m\angle SPT = (11x - 20)^\circ$, and $m\angle HTC = (8x + 5)^\circ$.



- a. Find the value of x .
 $x = 13$
- b. What is the measure of $\angle SPT$?
 123°
- c. What is the measure of $\angle HTC$?
 109°



Common Misconceptions

Students may understand the relationships of interior and exterior angles of triangles, but still result in an incorrect answer. Consider asking students to explain their thought process when solving to determine if an incorrect answer is a result of students struggling with the new material or a miscalculation in solving.



- How did you find the answer?
- Why did you choose this strategy?
- How can you check that your answers are correct?

11.5.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



11.5.5 Wrap-Up: Cool Down

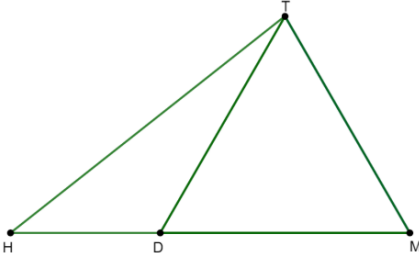
1. Write a summary of what you learned in this lesson.

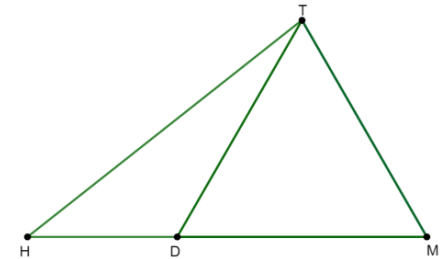
Answers will vary.

11.6 Triangle Inequality Theorem

Lesson Introduction

 Benchmark Focus	MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.		 Learning Targets	- I can determine if a set of triangle side lengths can be used to construct a triangle. - I can use the Triangle Inequality Theorem to prove relationships in triangles.
 Benchmark Coherence	Building On MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.			Working Toward N/A
 Essential Questions	- How do you determine if a set of side lengths can be used to construct a triangle? - How can you use the Triangle Inequality Theorem to prove relationships in triangles?			
 Lesson Narrative	In this lesson, students discover how to use the Triangle Inequality Theorem to determine if a set of triangle side lengths can be used to construct a triangle. Students explore how to apply the theorem to prove that the order of the side lengths is the same as the order of their opposite angle measurements.			
 Component Summary	11.6.1 Warm-Up (~5 min.)	Solve a number puzzle (<i>SE p. 208</i>)	11.6.4 Exploration (5 min.)	Measure interior angles in triangle; list angles and side lengths from least to greatest (<i>SE p. 210</i>)
	11.6.2 Exploration (10 min.)	Explore if triangles can be constructed with side lengths (<i>SE p. 208</i>)	11.6.5 Guided Instruction (10 min.)	Discover how to prove inequality in a triangle (<i>SE p. 211</i>)
	11.6.3 Try It (5 min.)	Practice applying the Triangle Inequality Theorem (<i>SE p. 210</i>)	11.6.6 Try It (5 min.)	Order angles from least to greatest (<i>SE p. 212</i>)
	11.6.7 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 213</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Triangle Inequality Theorem - Whole Greater Than Parts Theorem <u>Additional Glossary:</u> - N/A		- Rulers - Protractors - Pieces of Paper	Prior to the lesson, work through 11.6.2 Exploration, 11.6.4 Exploration, and 11.6.5 Guided Instruction to ensure that you are familiar with the steps students will take and are prepared to support them as they work.

 Teacher Tips	Common Misconceptions - Students may not understand that the sum of any two sides in a triangle must be greater than the third side. - Students may not understand that the order of the measurements of the angles is the same as the order of their opposite side lengths. - Students may struggle with not having a visual when determining if given sides lengths create a triangle.	Things to Consider Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the relationship between the side lengths in triangles.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display During 11.6.6 Try It, consider using the instructional routine “Poll the Class” for question 1 for immediate feedback. Ask the students to hold up a whiteboard or piece of paper with the smallest angle written on it. Continue this for the next two angles. As students share their answer, scan the class to review students’ responses, and consider asking students to share why they chose that answer.	MA.K12.MTR.1.1: Participate in Effortful Learning 11.6.2 and 11.6.4 Explorations provide students with the opportunity to collaborate with their peers as they explore the relationships among the side lengths and angle measurements in triangles. These discussions will help students build confidence and understanding with the learning targets. Encourage students to justify their claims and discuss whether they agree or disagree with their peers and why.
 Differentiate Instruction	Consider using additional visuals, such as GeoGebra, to help students better conceptualize the concept of the Triangle Inequality Theorem for visual and kinesthetic learners. Students will have the chance to draw their own conclusions and discuss their rationale with peers, providing an opportunity for auditory learners to verbalize their understanding. 	
Lesson Notes:		



11.6.1 Warm-Up: Number Riddle

Component Overview: Students solve a number puzzle by substituting operations in a number sentence to obtain a given value. Solving non-routine problems, such as this puzzle, can help students build strategic competence.



11.6.1 Warm-Up: Number Riddle

1. Use any of the four operations (+, −, ×, ÷) with the numbers shown to produce a result of 36.

$$7 \times 5 + 3 - 2 + 0$$

7, 0, 2, 3, 5



Encourage students to use the “guess, check, and revise” strategy to solve the puzzle. Remind students to keep track of their attempts, so that they do not repeat attempts that were incorrect. Prompt students to share what they learned from their mistakes, and how their mistakes helped them improve their guesses.

11.6.2 Exploration: Triangle Inequality Theorem

Component Overview: Students explore if a triangle can be constructed from given side lengths and create conjectures about how to determine if triangle side lengths can be used to construct a triangle.



11.6.2 Exploration: Triangle Inequality Theorem

1. With your partner, use a ruler to determine if the given measurements can be used to create a triangle. Then, complete the table.

Measurements in Centimeters (cm)	Can it create a triangle?	Explanation
3 cm, 6 cm, and 8 cm	☐ Yes ☐ No	All the sides connect
3 cm, 4 cm, and 10 cm	☐ Yes ☐ No	All the sides don't connect
8 cm, 11 cm, and 15 cm	☐ Yes ☐ No	All the sides connect
2 cm, 5 cm, and 8 cm	☐ Yes ☐ No	All the sides don't connect
4 cm, 5 cm, and 11 cm	☐ Yes ☐ No	All the sides don't connect

2. What similarities did you notice about the measurements that could create a triangle?
Answers will vary. When the sum of two sides is greater than the third, all the sides connect.
3. What similarities did you notice about the measurements that could not create a triangle?
Answers will vary. When the sum of two sides is less than the third, all the sides do not connect.
4. Make a conjecture about how to determine if side lengths can be used to construct a triangle if you cannot draw it.
Answers will vary. If the sum of any two side lengths is greater than the third, then a triangle can be constructed.



For this Exploration, allow students to engage with it with their partner as they create triangles using the given lengths. Provide students with rulers and paper to use throughout this Exploration. Observe and listen as students work with their partner and ask students to share their ideas with the class at the conclusion of this activity.



Collect and Display

Listen for, collect, and display student conjectures about the relationships among the side lengths in triangles. Then, engage students in a debrief discussion about the Triangle Inequality Theorem. Include the theorem on an anchor chart or Word Wall and encourage students to reference it in subsequent components.

11.6.2 Exploration: Triangle Inequality Theorem (continued)

5. Ask two other pairs of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of the Conjecture	Initials
<i>Answers will vary</i>	
<i>Answers will vary</i>	

6. Review your conjecture with your partner and if necessary, make any changes.
7. Use your conjecture to determine if 3 cm, 4 cm, and 7 cm will create a triangle.
No because $3 + 4$ is not greater than 7.
8. Check your answer by constructing the triangle with a ruler.
9. What did you notice about the measurements 3 cm, 4 cm, and 7 cm?
Two of the sides together were the exact length of the third side.
10. Adjust your conjecture if needed.



Consider using a dynamic tool, such as GeoGebra, to provide entry points for visual and kinesthetic students to investigate the relationship between the side lengths in triangles.

Suggested GeoGebra Link:
<https://www.geogebra.org/m/K5CEeBEu>



Triangle Inequality Theorem

The sum of the lengths of any two sides of a triangle is greater than the length of the third side.



If three segments have lengths such that the sum of any two is greater than the third, then a triangle can be constructed from the segments.

11.6.3 Try It: Triangle Inequality Theorem

Component Overview: Students practice applying the Triangle Inequality Theorem to determine if given side lengths can be used to construct a triangle.



11.6.3 Try It: Triangle Inequality Theorem

1. Complete the table.

Side Lengths	Work	Can it create a triangle?
57, 16, and 75	$57 + 16 < 75$ $57 + 75 > 16$ $75 + 16 > 57$	☐ Yes ☒ No
33, 12, and 25	$33 + 12 > 25$ $33 + 25 > 12$ $25 + 12 > 33$	☒ Yes ☐ No
21, 5, and 26	$5 + 21 = 26$ $5 + 26 > 21$ $26 + 21 > 5$	☐ Yes ☒ No
61, 100, and 61	$61 + 61 > 100$ $61 + 100 > 61$ $61 + 100 > 61$	☒ Yes ☐ No
48, 24, and 21	$21 + 24 < 48$ $21 + 48 > 24$ $48 + 24 > 21$	☐ Yes ☒ No



Provide a kinesthetic entry point by debriefing the activity with the following activity:

Designate one side of the room as “yes,” and the other side of the room as “no.” Read the side lengths aloud. Prompt students to walk to the side of the room that represents their answer. Ask students on each side of the room to justify their answers. Challenge students by including additional side lengths that they did not see in the component. The student with the most descriptive explanation of their answer could also be given the opportunity to create and call out the side lengths in the next round.



- How did you determine the answer?
- What mental math strategies did you use to calculate the sums?
- Did you test all of the possible pairs of sums of the side lengths?

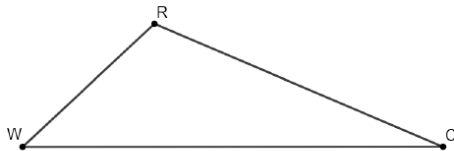
11.6.4 Exploration: Relating Angle Measures and Sides of Triangles

Component Overview: Students use protractors to measure the interior angles in a triangle. Then, they list the angle measurements and side lengths from least to greatest. Students discover that the order of the side lengths is the same as the order of their opposite angle measurements.



11.6.4 Exploration: Relating Angle Measures and Sides of Triangles

$\triangle RCW$ is shown, where $WC = 24$, $CR = 18$, and $RW = 10$.



1. Use a protractor to find the angle measures. Record your answers in the table.

Angle Measures	
$m\angle CWR =$	43°
$m\angle WRC =$	114°
$m\angle RCW =$	23°

2. List the side lengths in order from longest to shortest.
 WC, RC, WR
3. List the angles in order from largest to smallest.
 $\angle WRC, \angle CWR, \angle RCW$
4. What do you notice about the size of the angle compared to the length of the opposite side?
The longest side is opposite the larger angle and the shortest side is opposite the smaller angle.



Provide students the opportunity to create a conjecture about the size of the angle compared to the length of the opposite side. Allow students to share their conjecture with a partner to provide an entry point for auditory learners.

11.6.5 Guided Instruction: Proving Inequality in a Triangle

Component Overview: Students discover how to prove inequalities in a triangle. Engage students in discussions throughout the component to ensure that they understand the key terms and the proof.



11.6.5 Guided Instruction: Proving Inequality in a Triangle



If two **sides** of a triangle are unequal in measure, then the angle opposite the longer side has the greater measure.



If two **angles** of a triangle are unequal in measure, then the side opposite the angle with the greater measure is the longer side.

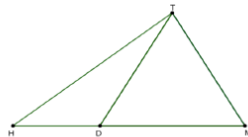


Whole Greater Than Parts Theorem

If $a > 0$, $b > 0$, and $a + b = c$, then $c > a$ and $c > b$.

$\triangle THM$ is shown, with $\overline{TD}$ creating $\triangle HTD$ and $\triangle DTM$, where $\triangle DTM$ is an equilateral triangle.

Given: $\triangle DTM$ is equilateral
Prove: $HM > HT$



Statement	Reason
1. $\triangle DTM$ is equilateral; $m\angle DTH = 22^\circ$	1. Given
2. $m\angle TMD = m\angle DTM$	2. Property of equilateral triangle
3. $m\angle DTM + m\angle DTH = m\angle MTH$	3. <i>Angle Addition Postulate</i>
4. $m\angle TMD + m\angle DTH = m\angle MTH$	4. Substitution Property of Equality
5. $m\angle MTH > m\angle TMD$	5. Whole greater than parts theorem
6. $HM > HT$	6. <i>If two angles of a triangle are unequal in measure, the side opposite the angle with the greater measure is the longer side.</i>



Add key terms and an additional glossary to the classroom Word Wall for students to reference in this and subsequent lessons.



- What do you know about the triangle based on the given information?
- How do the side lengths compare? How do the angle measurements compare?



Provide a visual entry point by prompting students to color code and label the diagrams as they work through each step of the proof.

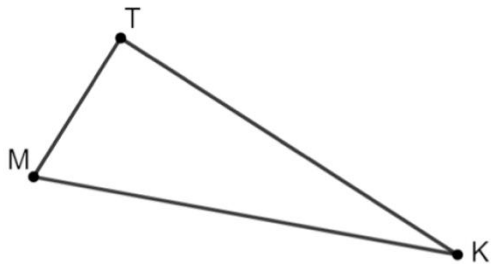
11.6.6 Try It: Triangle's Angle Measures

Component Overview: Students use the given triangle side lengths to order the angle measurements from least to greatest.



11.6.6 Try It: Triangle's Angle Measures

- Triangle TKM is shown, where $KT = 32$, $MK = 34$, and $TM = 13$. List the measures of the angles in order from least to greatest.



$m\angle TKM, m\angle TMK, m\angle MTK$



Poll the Class

Consider using the instructional routine “Poll the Class” for question 1 for immediate feedback. Ask the students to hold up a whiteboard or piece of paper with the smallest angle written on it. Continue this for the next two angles. As students share their answers, scan the class to review students’ responses, and consider asking students to share why they chose that answer.



What strategies did you use to help you determine your answer?

11.6.7 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



11.6.7 Wrap-Up: Cool Down

1. List three side lengths that would not make a triangle.
Explain how you know those sides would not make a triangle.

Answers will vary

2. List three side lengths that would make a triangle.
Explain how you know those sides would make a triangle.

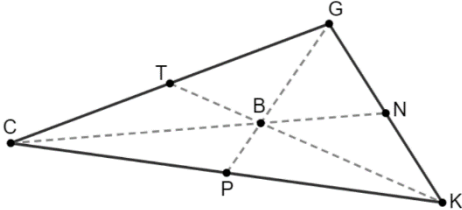
Answers will vary

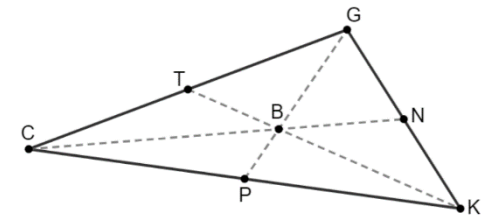
11.7 Centroid of a Triangle

Lesson Introduction

 Benchmark Focus	MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.		 Learning Targets	- I can find the centroid of a triangle. - I can use the relationship of the distance a centroid is from a vertex to the midpoint of the opposite side to solve problems.
 Benchmark Coherence	Building On MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.			Working Toward MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.
 Essential Questions	- How do you find the centroid of a triangle? - What is the relationship between the distance from the vertex to the centroid, and the distance from the centroid to the midpoint?			
 Lesson Narrative	In this lesson, students explore how to construct medians and centroids of triangles. Then, they discover that the length from the vertex to the centroid is twice the length from the centroid to the midpoint. Finally, they apply their knowledge of this relationship to solve problems.			
 Component Summary	11.7.1 Warm-Up (~5 min.)	Order angles from least to greatest based on given side lengths (<i>SE p. 214</i>)	11.7.3 Guided Instruction (10 min.)	Discover length from vertex to centroid is twice length from centroid to midpoint (<i>SE p. 217</i>)
	11.7.2 Exploration (20 min.)	Explore how to construct medians and centroids of triangles (<i>SE p. 214</i>)	11.7.4 Your Turn! (5-10 min.)	Practice solving problems involving medians and centroids; analyze error in sample work (<i>SE p. 218</i>)
	11.7.5 Wrap-Up (~5 min.)	Summarize lesson for hypothetical student who missed class (<i>SE p. 219</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Median of a Triangle - Centroid of a Triangle <u>Key Terms:</u> - Vertex - Midpoint		- Patty Paper - Compass - Triangle Manipulative (Optional) Colored Pencils	
		Additional Preparation Prior to the lesson, consider cutting out triangle manipulatives, made of cardboard or posterboard, to provide a hands-on approach to exploring the location of the centroid.		

 Teacher Tips	Common Misconceptions - Students may struggle with solving problems where the distance from the vertex to the centroid is twice the distance from the centroid to the midpoint. - Students may struggle with constructing perpendicular bisectors.	Things to Consider - Consider creating and displaying an anchor chart with the definition of median and centroid of a triangle or adding the terms to a classroom Word Wall. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the location of the centroid. - If time is an issue, consider assigning 11.7.4 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect During 11.7.2 Exploration, students have an opportunity to compare the distance from the vertex to the centroid to the distance from the centroid to the midpoint. After constructing the centroid, prompt students to mark each distance using patty paper. Ask students what relationship they notice and prompt them to test their conjectures with each median in the triangle.	MA.K12.MTR.4.1: Discussion and Reflection In 11.7.4 Your Turn!, students have the opportunity to demonstrate their mastery of the learning target by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors themselves in the future.
 Differentiate Instruction	Consider allowing students to find the balancing point in 11.7.2 Exploration by providing students with a triangular manipulative made from a piece of cardboard or poster board. Students can balance the triangular manipulative on their pencil to discover the location of the centroid. This provides students with visual and kinesthetic entry points. Encourage students to discuss their discoveries with a partner to provide entry points for auditory learners.	
Lesson Notes:		





11.7.1 Warm-Up: Bell Work

Component Overview: Students use the given triangle side lengths to order the angle measurements from least to greatest. This activity is based on the work that students completed in Lesson 6. Provide support to any students who struggle.



11.7.1 Warm-Up: Bell Work

- Given $\triangle BMT$, where $MB = 7$, $MT = 12$, and $BT = 9$, list the measures of the angles in order from least to greatest.
 $m\angle MTB, m\angle BMT, m\angle MBT$



Challenge students who finish early to create their own similar problems, swap problems with a peer, and solve each other's problems. Remind students to use the Triangle Inequality Theorem to make sure their side lengths in their problems can be used to construct a triangle.

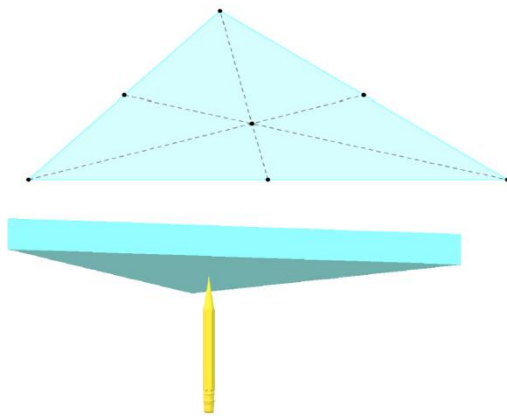
11.7.2 Exploration: Centroid of a Triangle

Component Overview: Students explore how to construct the median and the centroid of a triangle. Students have previously worked with medians and constructed perpendicular bisectors. However, this will be the first time students have been exposed to the centroid.



11.7.2 Exploration: Centroid of a Triangle

1. All polygons have a balancing point. The diagrams show the balancing point of a triangle.



- a. What do you notice?

Answers will vary. I notice that the balancing point is inside the triangle.

- b. Discuss with your partner how you think the balance point was found. Summarize your discussion.

Answers will vary. In the bottom picture it is where the triangle will balance on the pencil. In the top picture, it is the point of concurrency of the lines drawn from each vertex to the midpoint of the opposite side.

A **median of a triangle** is a segment that joins any vertex of the triangle to the midpoint of the opposite side.



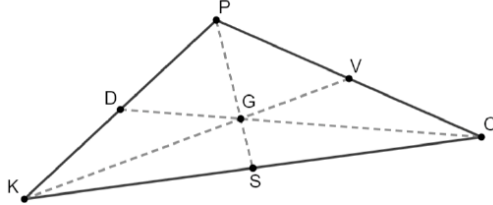
Provide kinesthetic and visual entry points by giving students triangular manipulatives (made from cardstock/cardboard or triangular snack chips, such as Doritos). Prompt students to find the balancing point with their pencils. Connect to students' prior experiences balancing on a see saw.



- *How did you find the balancing point?*
- *Where is the balancing point located in relation to the triangle's vertices and sides?*

11.7.2 Exploration: Centroid of a Triangle (continued)

2. $\triangle KPC$ is shown, where point G is a point of concurrency.



Discuss with your partner how to find the midpoints of the triangle's sides using each method. Complete the table.

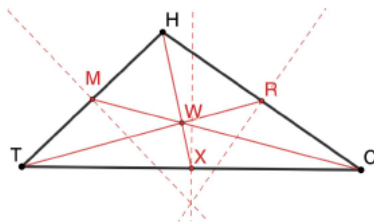
Patty Paper	Compass
Answers will vary. Fold each side so that two vertices line up. Crease the side at the midpoint. Repeat for each side.	Answers will vary. Use the compass to find each side's perpendicular bisector and the midpoint will be at the point of intersection.



- What are the similarities and differences between how you find the midpoints of triangles using patty paper versus using a compass?
- Which method do you prefer and why?

11.7.2 Exploration: Centroid of a Triangle (continued)

3. $\triangle HCT$ is shown.



- a. Use the steps to construct the centroid of $\triangle HCT$.
Construction shown above.

Step 1: Trace $\triangle HCT$ on patty paper.
Step 2: Construct the perpendicular bisector of $\overline{HT}$ to find its midpoint, M .
Step 3: Draw the median from M to the opposite vertex, $\angle HCT$.
Step 4: Construct the perpendicular bisector of $\overline{HC}$ to find its midpoint, R .
Step 5: Draw the median from R to the opposite vertex, $\angle HTC$.
Step 6: Construct the perpendicular bisector of $\overline{TC}$ to find its midpoint, X .
Step 7: Draw the median from midpoint X to the opposite vertex, $\angle THC$.
Step 8: Draw point W , the point of concurrency, where medians $\overline{RT}$, $\overline{XH}$, and $\overline{MC}$ intersect.

- b. What do you notice about the location of point W in $\triangle HCT$?

Answers will vary. Point W is on the inside of the triangle and located at the intersection of all the medians of triangle HCT .



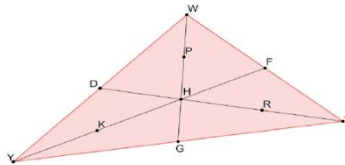
Consider creating an anchor chart with the steps required to construct a centroid. Add a third column with a visual diagram of each step of the construction. Color code and label the diagrams to provide students with a visual entry point.



Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the location of the centroid.

11.7.2 Exploration: Centroid of a Triangle (continued)

4. $\triangle YWT$ is shown, with medians $\overline{WG}$, $\overline{TD}$, and $\overline{YF}$. Point P is the midpoint of $\overline{WH}$, point R is the midpoint of $\overline{TH}$, and Point K is the midpoint of $\overline{YH}$.



- a. Copy the first segment using patty paper. Use the copy to determine the relationship the first segment and the point have with the second segment. Complete the table.

First Segment	Point	Second Segment	Relationship
$\overline{DH}$	R	$\overline{TH}$	$\overline{TH}$ is twice the length of $\overline{DH}$
$\overline{FH}$	K	$\overline{YH}$	$\overline{YH}$ is twice the length of $\overline{FH}$
$\overline{GH}$	P	$\overline{WH}$	$\overline{WH}$ is twice the length of $\overline{GH}$

The **centroid of a triangle** is the intersection of the three medians of the triangle (each median connecting a vertex with the midpoint of the opposite side).

- b. Write a conjecture about the distance a centroid is from the vertex and the distance a centroid is from the midpoint of the opposite side.
The centroid is double the distance from a vertex in comparison to its distance from the opposite side's midpoint.



Compare and Connect

Students have an opportunity to compare the distance from the vertex to the centroid to the distance from the centroid to the midpoint. After constructing the centroid, prompt students to mark each distance using patty paper. Ask students what relationship they notice and prompt them to test their conjectures with each median in the triangle.



Consider creating and displaying an anchor chart with the definition of median and centroid of a triangle, or adding the terms to a classroom Word Wall.

11.7.3 Guided Instruction: Centroid of a Triangle

Component Overview: Students discover that the length from the vertex to the centroid is twice the length from the centroid to the midpoint.



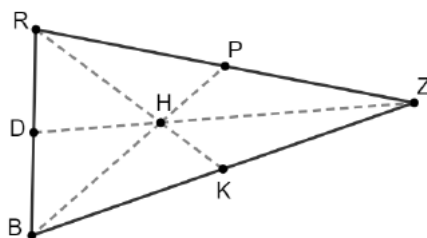
11.7.3 Guided Instruction: Centroid of a Triangle



The length from the vertex to the centroid is twice the length from the centroid to the midpoint of the opposite side, which yields a ratio of 2:1.

1. $\triangle RZB$ is shown, where point H is the centroid of the triangle.

$DH = 11.3$, $RK = 20.55$, and $BH = 15.2$.



- a. Complete the table.

$HZ = 22.6$	$DZ = 33.9$
$RH = 13.7$	$HK = 6.85$
$HP = 7.6$	$BP = 22.8$

- b. If $BK = 6x + 3$ and $KZ = 10x - 7$, what is the length of BZ ?

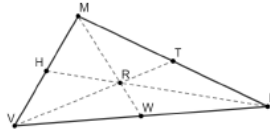
36



This component will consolidate the learning from 11.7.2 Exploration. Consider prompting students to use colored pencils to mark their diagrams while you read the text in the blue box aloud. For example, the segments from the vertex to the centroid could be highlighted one color, and the segments from the centroid could be highlighted in another color. Students could also label the segment from the centroid to the midpoint with a variable and the segment from the centroid to the vertex with two times the variable. Model how to find a few of the measurements in the table, and then release students to work on the remaining parts independently.

11.7.3 Guided Instruction: Centroid of a Triangle (continued)

2. $\triangle MPV$ is shown, where point R is the centroid of the triangle. $MR = 5x - 9$, $RP = 3w + 7$, $RV = 12y - 9$, $RW = 1.75x$, $RT = 4y + 2$, and $HR = w + 9$.



- a. Find the value of each variable.

$x = 6$	$y = 3.25$	$w = 11$
---------	------------	----------

- b. Complete the table.

$HR = 20$	$RP = 40$	$HP = 60$
$RT = 15$	$VR = 30$	$VT = 45$
$RW = 10.5$	$MR = 21$	$MW = 31.5$



Think-Pair-Share

Prompt students to initially solve the problems individually. Next, ask students to discuss how they obtained their answers with a partner. Finally, call on a variety of students to share their strategies with the whole class. Ask students if they agree or disagree with their peers' strategies and why.

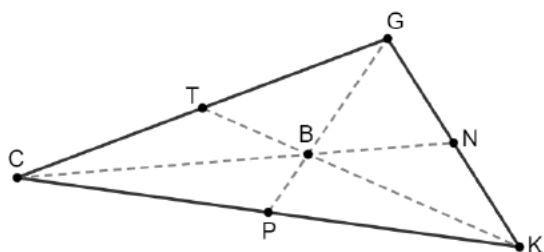
11.7.4 Your Turn!

Component Overview: Students practice solving problems involving medians and centroids. Students also analyze errors in a sample of student work.



11.7.4 Your Turn!

1. $\triangle CGK$ is shown, where point B is the centroid of the triangle. $TB = \frac{1}{2}x - 4$, $PB = 4.4$, $CB = y + 6$, $BK = 2x - 31$, and $BN = \frac{1}{3}y + 5$.



- a. What is the value of x ?

$$x = 23$$

- b. What is the value of y ?

$$y = 12$$

- c. Complete the table.

$TB = 7.5$	$BK = 15$
$CB = 18$	$CN = 27$
$BG = 8.8$	$PG = 13.2$

- d. If $CT = 4k - 10$ and $TG = 2k + 1$, what is the length of $\overline{CG}$?

$$24$$



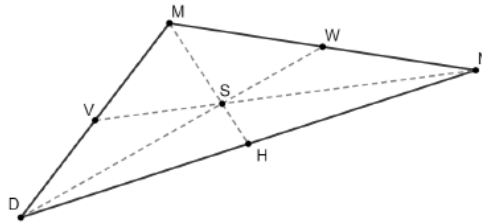
Provide a visual entry point by prompting students to color code and label their diagrams.



- How did you find the value of x and the value of y ?
- How can you check your work?

11.7.4 Your Turn! (continued)

2. $\triangle DMN$ is shown, where point S is the centroid.



- a. Leonardo is finding the value of x using the information $VS = \frac{1}{2}x + 1$ and $SN = 2x - 6$. His work is shown.

Leonardo's Work
$2(2x - 6) = \frac{1}{2}x + 1$ $4x - 12 = \frac{1}{2}x + 1$ $\frac{7}{2}x = 13$ $x = \frac{26}{7}$

- b. Write a note to Leonardo explaining his mistake.
Leonardo, $\overline{VN}$ is in a 1:2 ratio with $\overline{VS}$ being the shorter segment. To set up your equation, VS should be multiplied by 2 and set equal to SN .
- c. What is the value of x ?
 $x = 8$



Challenge students who finish early to research real-world applications of finding the centroid of a triangle. For example, determining a central location for facilities (i.e. a restroom, restaurant, pool), or finding the center of gravity of a plane or bus to know where to put the fuel and luggage.



Students have the opportunity to demonstrate their mastery of the learning target by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors themselves in the future.

11.7.5 Wrap-Up: Cool Down

Component Overview: Students write a note to a hypothetical absent student that summarizes the lesson learning targets.



11.7.5 Wrap-Up: Cool Down

1. Write a short note summarizing today's lesson to a friend who missed class.

Answers will vary.

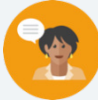
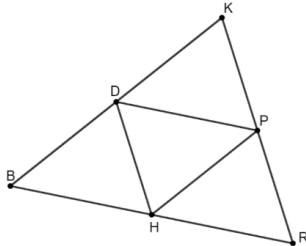


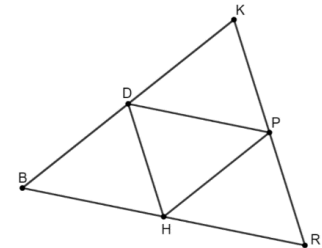
Providing students with sentence stems can help them compose their summary. Consider asking students to create a question to include in their summary.

11.8 Triangle Midsegment Theorem

Lesson Introduction

 Benchmark Focus	MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.		 Learning Targets	- I can discover the triangle midsegment theorem using AA similarity. - I can use the triangle midsegment theorem to prove relationships and to solve problems.
 Benchmark Coherence	Building On		Working Toward	
	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.		MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.	
 Essential Questions	- What is the relationship between the triangle midsegment theorem and AA similarity? - How can you use the triangle midsegment theorem to solve problems?			
 Lesson Narrative	In this lesson, students use angle-angle similarity to discover the triangle midsegment theorem. Then, students use the triangle midsegment theorem to prove relationships and solve problems.			
 Component Summary	11.8.1 Warm-Up (~5 min.)	Notice and Wonder about a sculpture of cubes (<i>SE p. 220</i>)	11.8.3 Guided Practice (10-15 min.)	Use triangle midsegment theorem to solve guided problems (<i>SE p. 222</i>)
	11.8.2 Exploration (20 min.)	Discover the triangle midsegment theorem (<i>SE p. 221</i>)	11.8.4 Your Turn! (5-10 min.)	Use triangle midsegment theorem to independently solve problems (<i>SE p. 224</i>)
	11.8.5 Wrap-Up (~5 min.)	Identify and correct errors in a sample of student work (<i>SE p. 224</i>)		
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> Triangle Midsegment Theorem <u>Additional Glossary:</u> N/A	(Optional) Colored Pencils	N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with identifying similar triangles by Angle-Angle Similarity. - Students may struggle with using the triangle midsegment theorem and may miscalculate when using it, resulting in an incorrect answer.	- Consider providing colored pencils or highlighters for students to color code and label their diagrams throughout this lesson. - Consider using a dynamic tool, such as GeoGebra, to enable students to investigate the location of the midsegment.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 11.8.1 Warm-Up, students are shown a photograph of a sculpture made of cubes. Ask students “What do you notice? What do you wonder?” Give students a few minutes to write down things they notice, and things they wonder. After students have had a chance to write down their responses, ask several students to share things they noticed and wondered. Record these for all to see. Steer the conversation to wondering about how the sculpture remains balanced.	MA.K12.MTR.4.1: Discussion and Reflection In 11.8.5 Wrap-Up, students have the opportunity to demonstrate their mastery of the learning targets by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors themselves in the future.
 Differentiate Instruction	Provide an auditory entry point in 11.8.2 Exploration by having students discuss what they notice about the relationship between the triangles. Provide a kinesthetic entry point by having students use Geoboards, Magformers, Busy Bags, or Exploragons to physically model the diagrams in 11.8.3 Guided Practice. Provide a visual entry point by prompting students to color code and label their diagrams in 11.8.4 Your Turn!	
		
Lesson Notes:		



11.8.1 Warm-Up: “Many Small Cubes”

Component Overview: Students share what they notice and wonder about a sculpture of cubes. This task is intended to inspire student curiosity and interest. This component can help build students’ agency and mathematical identities, as each and every student can participate.



11.8.1 Warm-Up: “Many Small Cubes”

1. Japanese architect Sou Fujimoto exhibited a sculpture of stacked and suspended aluminum cubes as a part of the art fair in the Parisian Jardins des Tuileries’ gardens.



- a. What do you notice about the structure?
Answers will vary. I notice that there are all kinds of different sized cubes. I notice that there are gaps between the cubes.
- b. What do you wonder?
Answers will vary. I wonder how they all stay stacked the way they are. I wonder if there is anything holding them together.



Notice and Wonder

Students are shown a photograph of a sculpture made of cubes. Ask students “What do you notice? What do you wonder?” Give students a few minutes to write down things they notice, and things they wonder. After students have had a chance to write down their responses, ask several students to share things they noticed and wondered. Record these for all to see. Steer the conversation to wondering about how the sculpture remains balanced.

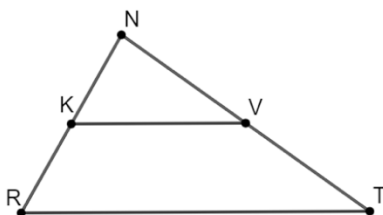
11.8.2 Exploration: Triangle Midsegment Theorem

Component Overview: Students use the fact that triangles have two congruent angles to prove that the triangles are similar. Then, students use the proportionality of the side lengths to determine missing side lengths in the two triangles. Finally, students discover that the midsegment is half the length of the third side, which it is parallel to.



11.8.2 Exploration: Triangle Midsegment Theorem

$\triangle NKV$ and $\triangle NRT$ are shown, where $\overline{KV} \parallel \overline{RT}$.



- Discuss with your partner what can be concluded from the given information. Justify your conclusions.
Answers will vary. Because the parallel lines create two sets of congruent corresponding angles, the two triangles are similar by AA similarity.
- Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials
<i>Answers will vary.</i>		
<i>Answers will vary.</i>		



Collect and Display

Listen for, collect, and display student conjectures about the relationship between the triangles. Display the Angle-Angle proof of triangle similarity on a classroom Word Wall or anchor chart.



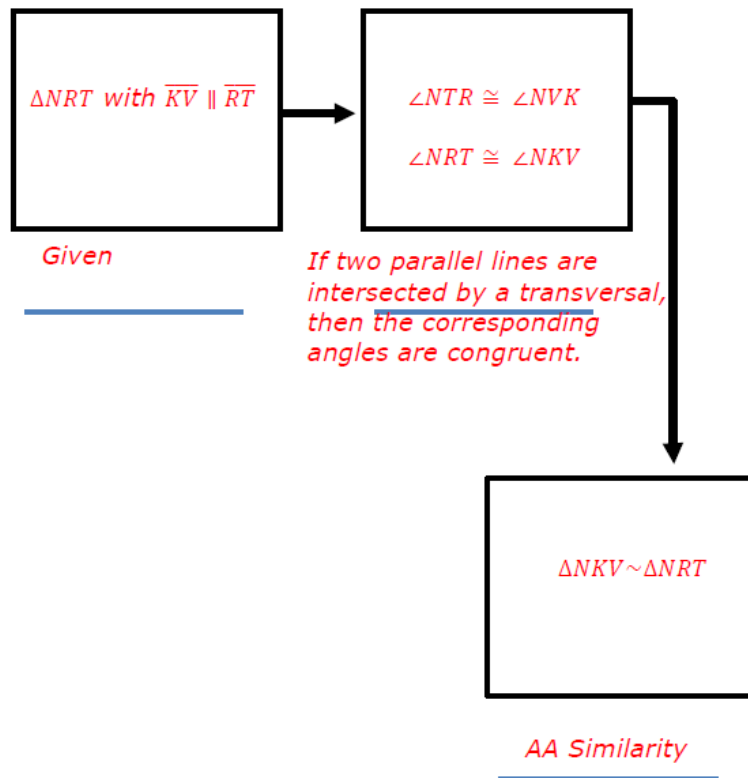
- What are your conclusions?
- Do you agree or disagree with your peers' conclusions? Why or why not?

11.8.2 Exploration: Triangle Midsegment Theorem (continued)

3. Given: $\triangle NRT$ with $\overline{KV} \parallel \overline{RT}$

Prove: $\triangle NKV \sim \triangle NRT$

Complete the flowchart proof.



Provide an auditory entry point by having students discuss what they notice about the relationship between the triangles. The flow chart proof provides students with a visual entry point. It also includes less words than a traditional proof, helping students comprehend the concept.

11.8.2 Exploration: Triangle Midsegment Theorem (continued)

4. What other information about triangles NKV and NRT can be determined from knowing that $\triangle NKV \sim \triangle NRT$?

Answers will vary. All of the corresponding sides will be proportional.

5. Complete the table using the information $KV = 5$, $NR = 6$, $NK = 3$, and $NT = 8.8$.

$RT = 10$	$NV = 4.4$
$KR = 3$	$VT = 4.4$

6. What do you notice about the location of $\overline{KV}$?
It connects the midpoints of $\overline{NR}$ and $\overline{NT}$.
7. What do you notice about the lengths of $\overline{KV}$ and $\overline{RT}$?
 $\overline{KV}$ is half the length of $\overline{RT}$.



Triangle Midsegment Theorem

The midsegment of a triangle joins the midpoints of two sides of a triangle such that its length is half the length of the third side of the triangle, and it is parallel to the third side of the triangle.



Consider having students label the triangle with the information given and found in question 5. This could provide students with a better understanding of the location and lengths by providing students with a visual.



Think-Pair-Share

Consider providing students with the opportunity to share with a partner what they notice after individual think time. Engaging the class in a group discussion will provide entry points for auditory learners.

11.8.3 Guided Practice: Triangle Midsegment Theorem

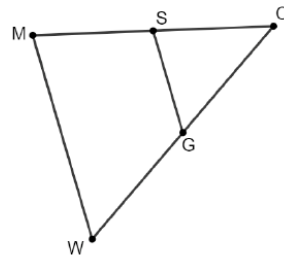
Component Overview: Students use the triangle midsegment theorem to solve multi-step problems. Model how to solve the problems and support students as they work during this component.



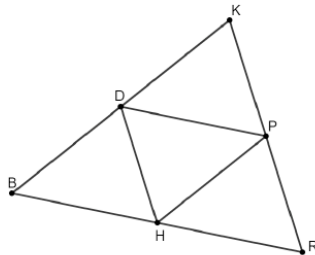
11.8.3 Guided Practice: Triangle Midsegment Theorem

1. $\triangle MCW$ is shown, where $\overline{SG}$ is the midsegment.

$$\begin{aligned} MS &= \left(\frac{1}{2}x + 4\right), \\ GC &= (11z - 2), \\ SG &= (2y + 5), \\ MW &= (8y - 2), \\ SC &= \left(\frac{3}{4}x - \frac{1}{4}\right), \\ WG &= (9z + 1). \end{aligned}$$



- Find the value of x .
 $x = 17$
 - Find the value of y .
 $y = 3$
 - Find the value of z .
 $z = 1.5$
 - What is the perimeter of $\triangle MCW$?
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2. $\triangle BKR$ is shown, where D , P , and H are the midpoints and the perimeter of $\triangle DPH$ is 23.5. $DP = 17x$, $PH = (6x + 5)$, and $HD = 20x - 3$.



- What is the value of x ?
 $x = 0.5$



For this Guided Practice, determine how much guidance is needed for your students based on observations during the 11.8.2 Exploration. If necessary, consider modeling how to solve question 1 to address any misconceptions students may have.



Think-Pair-Share

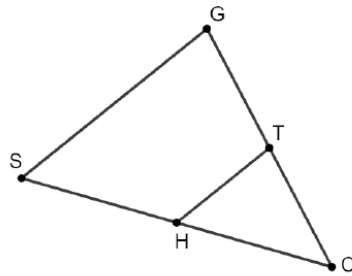
Prompt students to initially try the problems independently. Next, ask students to discuss their strategies and answers with a partner. Finally, call on a variety of students to share with the whole class during a debrief discussion. Ask other students if they agree or disagree and why.

11.8.3 Guided Practice: Triangle Midsegment Theorem (continued)

b. Complete the table.

$BK = 16$	$DH = 7$
$BR = 17$	$KP = 7$
$DP = 8.5$	$KR = 14$

3. $\triangle SGC$ and $\triangle HTC$ are shown.



Given: $SH = HC$ and $GT = TC$

Prove: $\overline{SG} \parallel \overline{HT}$

Since $SH = HC$ and $GT = TC$, point H is the midpoint of $\overline{SC}$ and point T is the midpoint of $\overline{GC}$ by the definition of midpoint. Therefore, $\overline{HT}$ is the midsegment of $\triangle SGC$ by definition and $\overline{SG} \parallel \overline{HT}$ by the triangle midsegment theorem.



Provide a kinesthetic entry point by having students use Geoboards, Magformers, Busy Bags, or Exploragons to physically model the diagram.



- How do you find the lengths of each line segment after finding the value of the variable?
- How can you check your work?



Consider providing students with a bank of words to choose from to complete the paragraph in question 3. Explain that words in the bank can be used more than once. Include words that are not used in the word bank. Prompt students to read and rewrite the paragraph several times to help them internalize the concepts.

11.8.4 Your Turn!

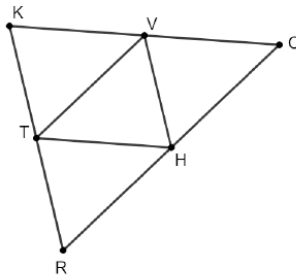
Component Overview: Students use the triangle midsegment theorem to independently solve problems. Circulate while students work, to provide scaffolds to struggling students and determine which student to call on during the debrief discussion.



11.8.4 Your Turn!

$\triangle KCR$ is shown, where T , V , and H are midpoints.

$KC = (5x)$, $VH = (\frac{1}{3}y + 11)$, $VT = (6w - 5)$, $KR = (2y - 2)$,
 $RC = (8w + 8)$, and $TH = (3x - 4)$.



- Find the value of each variable.

$x = 8$	$y = 18$	$w = 4.5$
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- Complete the table.

$TH = 20$	$KC = 40$
$VH = 17$	$KR = 34$
$VT = 22$	$RC = 44$



Clarify, Critique, Correct

Consider using the “Clarify, Critique, Correct” Instructional Routine for this activity. Have partners exchange their work to analyze, reflect on, and develop their partner’s work. This activity will help students improve upon their work by correcting errors and clarifying the meaning of their partner’s work. This routine fortifies output and engages students in meta-awareness.

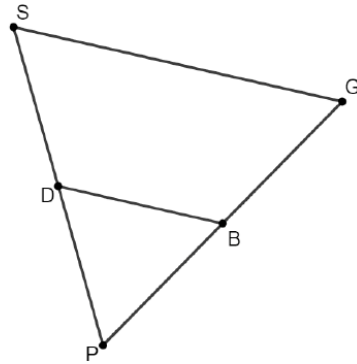
11.8.5 Wrap-Up: Error Analysis

Component Overview: Students identify and correct errors in a sample of student work.



11.8.5 Wrap-Up: Error Analysis

1. $\triangle SGP$ is shown, where $\overline{DB}$ is a midsegment. $SG = 18x + 22$ and $DB = 11x + 7$.



Priscilla was finding the value of x . Her work is shown.

Priscilla's Work
$18x + 22 = 11x + 7$
$7x = -15$
$x = -\frac{15}{7}$

- a. Write a note to Priscilla explaining her mistake.
Priscilla, when you set up your equation, you set the two lengths equal to each other, but the length of $\overline{DB}$ is half the length of $\overline{SG}$.
- b. Find the value of x .
 $x = 2$

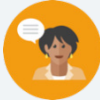


Students have the opportunity to demonstrate their mastery of the learning targets by identifying and correcting errors in a sample of student work. Discussing and reflecting on the sample student errors will decrease the likelihood of students making similar errors themselves in the future.

11.9 Solving Problems Using Triangle Relationships

Lesson Introduction

 Benchmark Focus	MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.		 Learning Target	I can solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.
 Benchmark Coherence	Building On MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.		Working Toward MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.	
 Essential Questions	How can you use what you know about triangle relationships, postulates, and theorems to solve mathematical and real-world problems?			
 Lesson Narrative	In this lesson, students consolidate and apply the knowledge and skills they obtained in the previous eight lessons. They use the triangle relationships, postulates, and theorems learned throughout the unit to solve mathematical and real-world problems.			
 Component Summary	11.9.1 Warm-Up (~5 min.)	Determine which coupon they would rather use (<i>SE p. 225</i>)	11.9.2 Activity (35-40 min.)	Apply triangle relationships, postulates, and theorems to solve problems (<i>SE p. 226</i>)
	11.9.3 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 227</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Triangle Inequality Theorem - Triangle Sum Theorem - Triangle Midsegment Theorem		- Copies of 11.9.2 Activity Questions (Optional) Colored Pencils	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may use the wrong relationship, postulate, or theorem when solving a problem. - Students may miscalculate when finding the value or a variable.	11.9.2 Activity is a mixed review of the concepts and procedures explored throughout the unit. Students will benefit from referencing relationships, postulates, and theorems on the classroom Word Wall or anchor charts.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Take Turns In 11.9.2 Activity, students apply relationships, postulates, and/or theorems that they have learned in this unit to solve mathematical and real-world problems. Consider having students work with a partner and take turns explaining how to solve the problem. Prompt students to discuss if they agree or disagree with their partners, and why.	MA.K12.MTR.3.1: Mathematical Fluency Students first explored the relationship, postulates, and theorems in the previous lessons in the unit. 11.9.2 Activity provides an additional opportunity for students to practice applying their knowledge and skills to solve mathematical and real-world problems. This additional practice will help students solidify their knowledge, skills, and confidence with the lesson learning targets from the unit.
	Provide a kinesthetic entry point by structuring 11.9.2 Activity as a circuit. Hang each problem with a different problem’s answer at different stations around the room. Students can start anywhere, but ask them to spread out, so no more than two students at each station. Have them work the problems against the walls or give them a book or clipboard to write on as they stand. Then, have students look around the room for the answer. When they find the answer, prompt them to solve the problem at that station. The activity is self-checking, because if students can complete the circuit, then they answered all of the problems correctly.	
Lesson Notes:		

11.9.1 Warm-Up: Would You Rather?

Component Overview: Students determine which coupon they would rather use and justify their decisions.



11.9.1 Warm-Up: Would You Rather?

1. Would you rather use a coupon worth ...

\$20 off your
entire purchase

OR

25% off your
entire purchase?

Answers will vary. It depends on how large the purchase is.

2. Explain your choice.

Answers will vary. Depending on how large the purchase is, I would choose the option that saves me the most money. Any item below \$80 is better to use the \$20 off. Any item above \$80 is better to use the 25% off.



This task should engage all students' interest because it is connected to saving money, a topic all consumers can benefit from understanding. The better coupon depends on the price of the item. Challenge students to determine the price when one coupon becomes more beneficial than the other.

11.9.2 Activity: Triangle Relationships

Component Overview: Students apply triangle relationships, postulates, and theorems to solve mathematical and real-world problems. Encourage students to show all of their work and check their answers.



11.9.2 Activity: Triangle Relationships

1. For each problem, record your work and answer in the table.

Problem Letter	Work	Answer
A	$9x - 15 + x + 11 + 3x + 2 = 180$ $x = 14$ $9(14) - 15 = 111$	111°
B	$5 + 5 = 10$ $10 < 11$	No, because $5 + 5 < 11$, meaning it does not meet the triangle inequality theorem.
C	$18x - 14 + 6x + 24 + 146 = 360$ $x = 8.5$ $18(8.5) - 14 = 139$	139°
D	$\frac{4}{3}x + 7 = \frac{1}{2}(8x + 2)$ $x = 2.25$ $\frac{4}{3}(2.25) + 7 = 10$	10 meters
E	$YN = 30, YW = 22, NW = 26$ $22 + 30 + 26 = 78$	78 units
F	$3x + 7 + 9x + 1 + 5x - 15 = 180$ $x = 11$ $9(11) + 1 = 100$	100°
G	$2(4x - 1) + 8x + 3 + 2\left(\frac{4}{3}x + 2\right)$ $= 84$ $x = 4.5$	4.5



Provide a kinesthetic entry point by structuring this component as a circuit. Hang each problem with a different problem's answer at different stations around the room. Students can start anywhere, but ask them to spread out, so no more than two students at each station. Have them work the problems against the walls or give them a book or clipboard to write on as they stand. Then, have students look around the room for the answer. When they find the answer, prompt them to solve the problem at that station. The activity is self-checking, because if students can complete the circuit, then they answered all of the problems correctly.



Take Turns

Consider having students work with a partner and take turns explaining how to solve the problem. Prompt students to discuss if they agree or disagree with their partners, and why.

11.9.3 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



11.9.3 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.